

# Promotion, Continuation, and Reputational Volatility under Task-Ease Uncertainty\*

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## Abstract

Should firms always promote after success? We show that promotion and continuation should be conditioned not on whether the outcome was good or bad, but on what it reveals about ability. In a two-period career-concerns model where success depends on managerial ability, task ease, and worker effort, the optimal promotion rule hinges on *reputational volatility*. With convex reputation utility, substitutability between ability and ease compresses volatility after success, making promotion after failure optimal; complementarity raises it, favoring promotion after success. The promotion rule also shapes project continuation: substitutability can justify continuing loss-making projects after failure. When a shelved project can still succeed, sidelining (promoting a worker while standing his project down) emerges as a graded reward after failure. Sidelining after success is never optimal both under substitutability and complementarity.

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## 1 Introduction

Promotion is usually thought of as a prize: it motivates effort, signals confidence to the labor market, and redeploys firm-specific human capital. But promoting a worker also does something else. By assigning him to manage a project, the firm also chooses where his ability will next be tested. Suppose the worker cares about his *reputation* as a manager. He therefore cares about which project he gets, because projects differ in how much they reveal about that ability. The design problem is sharpest in one case: promotion onto the very project he has just worked on. The worker can affect the project's outcome, and so he shapes what that outcome reveals about the project. When he is

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then promoted to run it, this project information becomes entangled with the market’s assessment of his own ability. The prize is thus shaped by the very performance the firm is trying to elicit. This paper asks when promotion is in fact good for the worker, and how a firm should award it.

In our setting, that information is how easy the task is.<sup>1</sup> The outcome cannot separate task ease from the incumbent manager’s ability: both raise the chance of success, and the market sees only the result. The worker first exerts effort under the incumbent. This effort is observable, so the outcome shifts the market’s belief about task ease only net of the effort the firm sees.<sup>2</sup> By the time he is promoted to run the project himself, the incumbent has departed and his own ability is untested. The belief about task ease is then the only thing the promotion carries forward, and it determines how strongly his future performance will move the market’s assessment of his ability.

Whether the first-period outcome, success or failure, amplifies or compresses what his future performance will reveal therefore determines how strong a prize promotion is, and turns the promotion rule into a design problem, one whose solution is governed by how the worker’s ability and the task’s ease combine to produce success: whether the two are complements or substitutes.

Our main result is that the value of internal promotion to the worker derives from the *volatility* of the reputation it exposes him to. We assume that the worker has convex utility over reputation, and therefore he prefers a more volatile assignment,<sup>3</sup> so the firm optimally promotes him onto whichever project his track record has made more volatile. Whether that is the project he just succeeded or just failed on is pinned down by a single structural feature: the modularity of ability and task ease in the success technology.<sup>4,5</sup> Under complementarity, the conventional prescription works: success

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<sup>1</sup>By “task ease” we mean a persistent, project-level factor, constant over the evaluation horizon: the latent market fit of a product line in sales, roster–system compatibility for a sports team, or the tractability of a research agenda in a lab, rather than a transitory shock.

<sup>2</sup>This is distinct from the ratchet effect, where an agent’s current performance, too, can shift beliefs about how hard his task is. There the channel is hidden action: the agent withholds effort to look less productive, and the revised belief tightens the terms the principal later imposes. With observable effort there is no such concealment; and the belief about task ease matters not because the firm can use it to demand more of the worker later, but because it changes how *informative* his next performance will be about his ability.

<sup>3</sup>Convexity is natural when the value of reputation is option-like: a worker captures the upside of a high assessment through outside offers and future wages while being insured on the downside. Holmström and Ricart i Costa (1986) show that the value of a manager’s human capital is option-shaped, hence convex, in perceived ability. In a different context, Li (2007) derive convex reputation payoffs in a related reporting environment. Section 2 gives an explicit option-value example.

<sup>4</sup>Substitutability (complementarity) corresponds to submodularity (supermodularity) of the success probability in ease and ability; see Section 2, especially Footnote 8.

<sup>5</sup>The notion of complementarity and substitutability in our model refers to the modularity of the success technology in ability and task ease, and is therefore distinct from the substitutability of effort across multiple tasks studied in the

raises volatility and the optimal rule rewards success. In contrast, under substitutability, *failure* raises volatility and the firm optimally promotes after failure (Proposition 1). The characterization imposes no functional form on reputation utility beyond convexity.

The mechanism is how perceived difficulty makes an outcome diagnostic of ability. Under complementarity, success needs both high ability and an easy task, so a hard-looking project makes outcomes mostly noise while an easy-looking one lets ability show; success then raises perceived ease, *widens* the future ability gap, and hands a worker promoted after success a high-volatility lottery. Under substitutability the logic inverts: either input suffices, so only a hard-looking project separates the able, and failure, by making the task look hard, *raises* volatility, so promotion after failure becomes the reward. In both regimes the worker's record reshapes the belief about the task, and modularity dictates whether that reshaping amplifies or compresses what future performance reveals.

Because the firm controls not only whom it promotes but whether the promoted project is actually run, the promotion rule reaches into the project-continuation decision. While running a project at full effort activates the reputational lottery, standing it down collapses the worker's payoff to his outside reputation. Under substitutability, the firm may *continue* the perceived-hard post-failure project to preserve the lottery that motivates first-period effort, even when the project is expected to lose money (Proposition 1). Under complementarity the firm does not promote after failure, and continuation reduces to a pure profitability decision.

*Sidelining* sharpens the picture: the firm promotes a worker onto a project but then leaves it idle. It matters most when the firm cannot decline to promote, as under seniority systems with strong job security, where the choice is which project to assign, not whether to promote (Section 5.1). A sidelined project delivers no reputational gain, only the worker's prior standing, so sidelining serves as a disciplinary device, optimal even when the project would be profitable to run; promotion can thus be reward or punishment, depending on the technology. Sidelining has a milder use as well (Section 5.2): when a shelved project still has some chance of success, it becomes a graded reward, between full continuation and no promotion, which under substitutability the firm grants after failure as a cheaper substitute for loss-making continuation. It never occurs under complementarity (Proposition 3) or after success (Proposition 4).

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multitask principal-agent literature (Holmström and Milgrom, 1991).

In sum, the optimal promotion system turns on whether ability and task ease are complements or substitutes in producing success. Which regime applies is a question about the environment, not about the model. Section 1.1 illustrates the two cases with drug development and sales territories.

Relative to promotions-as-incentives and career-concerns models, the value of promotion here is not a fixed prize: it is endogenous to post-promotion reputation dynamics, which is what lets promotion punish, and the informativeness of future performance is itself chosen by the firm through assignment, under convex, option-like reputation utility. Multidimensional-learning models have long recognized that what an outcome reveals about ability depends on beliefs about a confounding dimension; our contribution is the link built on it, that posterior dispersion inherits the sign of the complementarity between ability and task ease. A single structural primitive thus governs the second-moment dynamics of reputation, and through them the promotion rule, the continuation decision, and sidelining. Beyond convexity, no assumption on reputation utility is needed for the promotion and continuation results, and Section 5 characterizes how far the same holds for sidelining; Section 1.2 develops the connections to prior work.

The rest of the paper proceeds as follows. Section 1.1 develops the motivating examples and Section 1.2 the related literature. Section 2 presents the model and Section 3 the properties of reputation concerns. Section 4 characterizes the optimal contract and states the main result. Section 5 treats extensions (the seniority constraint, imperfect sabotage, unobservable effort, and intermediate modularity) and Section 6 concludes.

## 1.1 Motivating Examples

We illustrate how our framework maps onto concrete settings through two parallel cases corresponding to the complementarity and substitutability regimes. In each, a junior employee works under a senior manager on a unit of the firm; a first-period outcome is observed; the senior manager then exits and a leadership vacancy opens; and the firm chooses whether to elevate the junior employee to lead the same unit or to assign her to lead a different one. The cases differ only in how ability and task ease combine to produce success.

**Complementarity: drug-development programs.** Consider a pharmaceutical firm that runs several drug-development programs in parallel. An associate scientist has spent several years on a

program targeting a particular biological mechanism, working under a senior principal investigator, and early trial outcomes such as initial efficacy readouts are observed. The senior investigator then exits and a leadership vacancy opens. The firm chooses whether to promote the associate to lead the same program (continuing on the project whose target tractability has just been reassessed) or to assign her to a different program in the pipeline. Drug-development success requires both inputs at once: the target must be biologically tractable, and the scientist must possess the skill to develop it. Neither suffices alone: a highly skilled scientist cannot rescue an intractable target, and a tractable target still requires skill to advance. This is the hallmark of complementarity. That target tractability is a first-order and ex ante uncertain input is well documented: roughly ninety percent of candidates entering clinical development fail, and lack of efficacy (i.e., an intractable target or an incorrect biological hypothesis) is the single largest cause of attrition (Kola and Landis, 2004; Harrison, 2016). In such settings, success provides a relatively clean signal of the joint presence of both inputs and motivates assignment to the now-validated program.

**Substitutability: sales territories.** Consider a consumer-products firm that divides its market into regional territories, each with a designated account manager working under a regional sales director. After two years, territory revenue is observed; the director departs; and a managerial vacancy opens. The firm chooses whether to elevate the account manager to lead the same territory or to assign her to a different one. Territorial sales success is driven by two largely independent factors: the territory's intrinsic potential (demographics, competitive landscape, distribution access) and the salesperson's relationship-building and persuasion skill. Either alone can carry a territory to target: a favorable territory hits its numbers under a mediocre manager, and a skilled manager extracts performance from a difficult one. This is the hallmark of substitutability. The sales-force literature has long treated these as distinct dimensions. In Raju and Srinivasan (1996), salespersons differ in effectiveness while territories differ in their sales potential, so raw revenue confounds the two, and separating a manager's contribution from territory conditions requires explicitly accounting for those conditions.

In such settings success is ambiguous while failure points more strongly to a difficult environment, since either factor alone could have produced success. This asymmetry, as we show, reverses the optimal promotion rule.

These examples share the structural feature that our model captures: the period-1 outcome is informative about the project's task ease, while the (newly promoted) manager's ability remains to be tested in period 2. The firm's assignment decision is fundamentally about which project should serve as that test.

## 1.2 Related Literature

Our analysis connects three strands: promotions as an incentive device, career concerns and reputation, and learning under multidimensional uncertainty. The unifying theme is that the modularity of ability and task ease in the production technology shapes the informational content of performance signals and, through the value of the resulting reputational lottery, governs optimal promotion design.

**Promotion and Internal Labor Market** Beginning with Lazear and Rosen (1981), a large literature treats promotions as prizes in rank-order tournaments that motivate effort (Chan, 1996; Waldman, 2003, 2013); a parallel line studies the signaling role of promotions in conveying information to the labor market and shaping human-capital investment and turnover (Waldman, 1984; Ghosh and Waldman, 2010). A complementary strand grounds these analyses in the empirical regularities of internal labor markets, where promotions occur within stable hierarchical structures, are tied to wage dynamics, and reflect both incentive and assignment considerations (Baker, Gibbs, and Holmström, 1994a,b; Gibbons and Waldman, 1999).

Ricart i Costa (1988) and Martinez (2009) analyze the assignment-theoretic side of promotions, where the firm's promotion decision is shaped by what it learns about the worker. In nearly all of this work, however, promotion is treated as a reward whose value to the worker is exogenous. We instead endogenize the value of promotion through post-promotion reputation dynamics. This shows when promotion serves as a prize, and possibly as a punishment: a commitment to a low-volatility, reputation-poor assignment. This rationalizes promotion after failure and the use of project termination as a disciplinary device.

**Career concerns.** We build on Holmström (1999), in which implicit incentives arise because workers care about the market's assessment of their ability. Relative to the canonical model with one-dimensional ability uncertainty and exogenous signal informativeness, we add a second dimension

(task ease), so that the informativeness of future performance is itself an object the firm shapes through assignment. We also depart from the usual assumption of reputational risk aversion by allowing convex, option-like reputation utility, a reduced-form way to capture winner-take-all returns. Holmström and Ricart i Costa (1986) provide an early foundation, showing that the value of a manager’s human capital is option-shaped, hence convex, in perceived ability, and Li (2007) obtain convex reputation payoffs in a related reporting environment.

**Learning under multidimensional uncertainty.** Several studies share a structure with ours: learning about ability from output is confounded by a matching partner’s reputation, and players have incentives to move that confounding reputation. They do so through the choice of whom to match with in Anderson and Smith (2010), and through effort that shifts a teammate’s perceived ability, restoring the agent’s own commitment power, in Bar-Isaac (2007). Three things differ here: the confound is the task itself rather than a separate party; the worker is reassigned onto the very task his own outcome has moved; and what matters is the second moment, since posterior dispersion<sup>6</sup> inherits the sign of the ability–ease complementarity. Modularity thus governs the promotion rule, continuation, and sidelining.

In more recent literature on dynamic principal–agent setting, Fan and Johnson (2023) study dynamic incentives with substitutable shocks and risk-averse agents. Fan and Yang (2019) also study two-dimensional learning and show strategic suppression of a rival’s promotion when performances are substitutes; their focus is strategic interaction across agents, whereas ours is how the technology interacts with an individual’s career concerns. Closest in spirit is Loss and Renucci (2021), in which managers choose projects according to how those projects affect the dispersion and precision of posterior beliefs about their talent. While in their model the informativeness of a project is an exogenous attribute that the agent selects, our contribution is to endogenize it. We show that the reputational volatility of an assignment is governed by the modularity of the production technology: a past outcome shifts the perceived ease of the task, and modularity fixes whether higher perceived

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<sup>6</sup>The contrast with Anderson and Smith (2010) is worth making precise, since the mechanism by which dispersion creates value is shared. There the planner’s value rises in a mean-preserving spread of the *population* distribution of reputations, moved through the matching allocation, whereas here it is the posterior dispersion of a *single* worker’s reputation, moved through his own outcome. More substantively, in their model the informational value of a spread and the productive complementarity are separate forces whose conflict drives the failure of assortative matching, and the sign is assumed; here the sign of the dispersion is not a separate primitive but is inherited from the complementarity, so that modularity determines which outcome is volatile.

ease sharpens or blurs the ability signal carried by the next outcome. Volatility, the dispersion of beliefs rather than their level, thus becomes a structural object the firm assigns over, and the firm, not the agent, controls the assignment.

## 2 The Model

Consider a two-period model:  $t \in \{1, 2\}$ . A firm hires one worker and one manager each period. The worker and manager engage in one project. We assume that the project's outcome in period  $t$  is either success or failure, denoted by  $o_t \in \{s, f\}$ .

The project's success probability is determined by (i) the period- $t$  manager's managerial ability  $a_t \in \{H, L\}$  (high or low), (ii) the worker's effort  $e_t \in \{1, 0\}$ , and (iii) the task ease (manageability), denoted by  $d \in \{E, D\}$  (easy or difficult). The cost of effort is  $ce_t$  ( $c > 0$ ). The prior probability that any given manager has high ability is  $p_H = \Pr(a_t = H)$ . Abilities are drawn independently for each new manager. In contrast, the task ease  $d$  is constant throughout the horizon. The prior probability that the project is easy is denoted by  $\Pr(d = E) = p_E^0 \in (0, 1)$ . We assume that  $d$  and  $a_t$  are unobservable for all agents.

The probability of the project's success is given by  $q_{a_t d} \times e_t$ . We assume that  $q_{aE} \geq q_{aD} \geq 0$  and  $q_{Hd} \geq q_{Ld} \geq 0$  for each  $a \in \{H, L\}$  and  $d \in \{E, D\}$ . Define  $\Delta_H = q_{HE} - q_{HD}$ ,  $\Delta_L = q_{LE} - q_{LD}$ ,  $Q_H(p_E) = p_E q_{HE} + (1 - p_E) q_{HD}$ ,  $Q_L(p_E) = p_E q_{LE} + (1 - p_E) q_{LD}$ ,  $Q(p_E) = p_H Q_H(p_E) + (1 - p_H) Q_L(p_E)$ .<sup>7</sup>

We refer to  $\Delta_L / \Delta_H$  as the *degree of substitutability between ability and task ease*.<sup>8</sup> For example, when  $\Delta_L$  is small but  $\Delta_H$  is large (e.g.,  $q_{HE} > q_{HD} = q_{LE} = q_{LD}$ ), neither task ease nor ability alone substantially increases the probability of success. In contrast, when  $\Delta_L$  is large but  $\Delta_H$  is small (e.g.,  $q_{HE} = q_{HD} = q_{LE} > q_{LD}$ ), either task ease or ability can independently enhance the success probability.

Let  $\Pi$  be the profit from the project's success. We assume that  $Q(p_E^0)\Pi > c$  to focus on where implementing the project is optimal.

<sup>7</sup>We write  $Q_H$  or  $Q_L$  when the value of  $p_E$  is clear from the context or when the dependence on it is not essential.

<sup>8</sup>Formally, let  $(d, a) \in \{0, 1\}^2$ , where  $d = 1$  indicates an easy project and  $a = 1$  indicates a high-ability manager. Then, if  $\Delta_L / \Delta_H > 1$  (resp.,  $< 1$ ), the success probability exhibits submodularity (resp., supermodularity) with respect to task ease and ability.



Each worker's payoff consists of (i) an incentive fee  $\beta_t$  as a worker and (ii) the reputation for his ability  $a$  when he works as a manager. We assume that effort is observable to all parties but non-verifiable, so it cannot be contracted upon directly. The observability assumption lets us isolate reputational incentives from standard moral hazard; we relax this assumption in an extension (section 5.4.1). In contrast, we assume that the project's outcome  $o_t$  is verifiable and therefore contractible.

The reputation for managing ability is the probability that  $a = H$ , conditional on the worker's effort  $e_t$ , the public belief  $p_E$ , and the project's outcome  $o_t$ . Let  $P_H^{o_t}(p_E, e_t)$  denote the manager's reputation after the period- $t$  project outcome is known, when the public belief about project ease at the beginning of period  $t$  is  $p_E$  (i.e.,  $P_H^{o_t}(p'_E, e_t) = \Pr(a_t = H \mid e_t, o_t)_{|p_E=p'_E}$ ). Due to the observability of  $e$ , the public belief depends on  $e$ . We assume that the utility from reputation is  $v(P_H^{o_t}(p_E, e_t))$ , where  $v: [0, 1] \rightarrow \mathbb{R}$  is convex and strictly increasing. We also assume that  $v$  is not linear on any open interval containing  $p_H$ .

The convexity is a natural assumption. We provide an example that satisfies the convexity condition. Suppose that a worker engages in a new project with prior belief  $p_E$  about task ease after serving as the manager of the current project. Suppose also that there are many workers with a reputation  $p_H$ , which implies that the firm can employ such workers at low cost. To simplify, assume that the firm can employ workers with a reputation  $p_H$  at cost 0. Then, the firm needs to pay the worker with a reputation  $p'_H$  for the difference in the expected profit, which is calculated as

$$v(p'_H) = \max\{(p'_H - p_H), 0\} \cdot \Pi \cdot [p_E \cdot (q_{HE} - q_{LE}) + (1 - p_E) \cdot (q_{HD} - q_{LD})]. \quad (1)$$

This function is convex and non-linear around  $p_H$ .<sup>9</sup>

We suppose that the firm adopts internal promotion or external hiring. Let  $\pi_t^{o_t} \in [0, 1]$  be the probability of internal promotion.

The timeline is as follows. We assume that future payoffs are not discounted.

**Period 0.** The firm commits to a contract  $(\beta_1, (\beta_2^{o_1}, \pi_2^{o_1})_{o_1 \in \{s, f\}})$ .

**Period 1.** An incumbent manager (ability  $a_1$ ) and a worker are assigned to the project.

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<sup>9</sup>In a different context, Li (2007) shows that workers have convex utility functions about their reputation using a similar logic.

- (a) The worker chooses effort  $e_1 \in \{0, 1\}$ .
- (b) The project outcome  $o_1 \in \{s, f\}$  is realized; beliefs about  $d$  and  $a_1$  are updated.

**Period 2.** The incumbent manager exits. Based on  $o_1$ , the firm offers the period-1 worker for the manager of the current project with probability  $\pi^{o_1}$ . The period-1 worker has an outside option of not being a manager, and gets reputation payoff  $v(p_H)$ . If the period-1 worker is not assigned to the current project, the firm hires an external manager. A new worker is assigned to the project.

- (a) The new worker chooses effort  $e_2^{o_1} \in \{0, 1\}$ .
- (b) The project outcome  $o_2 \in \{s, f\}$  is realized; the period-2 manager's reputation is determined. The project ends.

### 3 Basic Properties of Reputation Concerns

This section summarizes the basic properties of reputation concerns. Given the outcome  $o_t \in \{s, f\}$  and possible effort realization  $e_1$ , under the current belief about the difficulty of the project,  $p_E$ , outsiders update their belief about the difficulty of the project  $P_E^{o_t}(p_E, e)$  and that about the current manager's ability,  $P_H^{o_t}(p_E, e)$ . These probabilities have the following properties.

**Lemma 1.** *For each  $e$  and  $p_E$ ,*

- (i)  $P_E^s(p_E, e)$  is independent of  $e$ .
- (ii)  $P_E^s(p_E, e) > p_E > P_E^f(p_E, e)$ .
- (iii)  $P_E^s$  and  $P_E^f$  are increasing in  $p_E$ .
- (iv)  $P_H^s(p_E, e) > p_H > P_H^f(p_E, e)$ .
- (v)  $E_o[P_H^o(p_E, e)] = p_H$ .
- (vi) (a) If  $\frac{\Delta_L}{\Delta_H} < \frac{q_{LE}}{q_{HE}}$ ,  $P_H^s(p_E, e)$  is increasing and  $P_H^f(p_E, e)$  is decreasing in  $p_E$ .  
 (b) If  $\frac{\Delta_L}{\Delta_H} > \frac{1-q_{LE}}{1-q_{HE}}$ ,  $P_H^s(p_E, e)$  is decreasing and  $P_H^f(p_E, e)$  is increasing in  $p_E$ .

Below, we explain the implications of Lemma 1. Properties (i)–(v) are standard: success raises perceived ease and ability, failure lowers both, and the posterior mean of ability equals the prior by the law of iterated expectations. From Lemma 1(i),  $P_E^s(p_E, e)$  is independent of  $e$ . We write  $P_E^s(p_E) = P_E^s(p_E, e)$  for brevity.

Lemma 1 (vi) shows the key property of our model. A low value of  $\frac{\Delta_L}{\Delta_H}$  implies complementarity between project ease and the manager's ability. In this case, conditional on project success, individuals tend to infer that the manager is more competent when the project appears easier. However, when  $p_E$  is low, even if managers have high ability, the overall success probability is also low due to the complementarity. As a result, differences in ability are less likely to explain the observed outcomes. In other words, observed success and failure are more likely to be attributed to luck rather than ability. This property implies that the reputational gap between success and failure becomes larger as  $p_E$  increases; when success is more informative (i.e., less likely to occur by chance), it more strongly influences how the manager is perceived. Under high  $\frac{\Delta_L}{\Delta_H}$  (substitutability), the comparative statics reverse: the reputational gap between success and failure *shrinks* as  $p_E$  increases.

We refer to *strong complementarity* as  $\frac{\Delta_L}{\Delta_H} < \frac{q_{LE}}{q_{HE}}$  and *strong substitutability* as  $\frac{\Delta_L}{\Delta_H} > \frac{1-q_{LE}}{1-q_{HE}}$ . The intermediate region is discussed in Section 5.4.2.

Using these results, we discuss the manager's utility from reputation, denoted by  $U$ . We define  $U$  as follows.

$$U(p_E, e) = Q(p_E) \cdot e \cdot v(P_H^s(p_E, e)) + (1 - Q(p_E) \cdot e) \cdot v(P_H^f(p_E, e))$$

The following lemma is immediate from Lemma 1 and the convexity of  $v$ .

**Lemma 2.** (i)  $U(p_E, 0) = v(p_H) < U(p_E, 1)$ .

(ii) If  $\Delta_L/\Delta_H < \frac{q_{LE}}{q_{HE}}$ ,  $U(p_E, 1)$  is increasing in  $p_E$ .

(iii) If  $\Delta_L/\Delta_H > \frac{1-q_{LE}}{1-q_{HE}}$ ,  $U(p_E, 1)$  is decreasing in  $p_E$ .

We write  $U(p_E) = U(p_E, 1)$  hereafter. Using these terms, the next section characterizes the optimal contract.

## 4 The optimal contract

This section characterizes the optimal complete contract that implements the full effort profile  $(e_1, e_2^s, e_2^f)$ . Here,  $e_2^{o_1}$  denotes the effort exerted by the second-period worker, conditional on the first-period project outcome  $o_1 \in \{s, f\}$ . Note that when the firm does not incentivize the second-period worker to exert effort, the project fails with certainty (since  $q_{ad} \times 0 = 0$ ). We refer to the case  $e_2^{o_1} = 0$  as *project sidelining*. The contract offered to managers and workers is initially defined as the tuple  $(\pi^s, \pi^f, \beta_2^s, \beta_2^f, \beta_1)$ . However, rather than focusing on the second-period incentive payments  $(\beta_2^s, \beta_2^f)$ , it is more transparent to analyze the contract in terms of the implemented effort profile  $(e_2^s, e_2^f)$ . Accordingly, we refer to  $(\pi^s, \pi^f, e_2^s, e_2^f, \beta_1)$  as the contract.

Throughout this section, let  $P_E^s = P_E^s(p_E^0, e = 1)$ . Note that  $P_E^f(p_E^0, e = 0) = p_E^0$ . When the firm offers a complete contract to employees,<sup>10</sup> the incentive compatibility condition for the period-1 worker to exert effort is the following:

$$Q(p_E^0)[\beta_1 + \pi^s U(P_E^s, e_2^s) + (1 - \pi^s)v(p_H)] + (1 - Q(p_E^0))[\pi^f U(P_E^f, e_2^f) + (1 - \pi^f)v(p_H)] - c \geq \pi^f U(p_E^0, e_2^f) + (1 - \pi^f)v(p_H). \quad (2)$$

Furthermore, as the period-1 worker can decline the offer and be a worker, individually rational condition  $U(p, e_2^{o_1}) \geq v(p_H)$  should be satisfied. However, this inequality is satisfied by the convexity of  $v$ .

Inequality (2) can be rearranged as

$$\beta_1 \geq B := \frac{1}{Q(p_E^0)} \left[ c - Q(p_E^0) \left[ \pi^s [U(P_E^s, e_2^s) - v(p_H)] - \pi^f [U(P_E^f, e_2^f) - v(p_H)] \right] + \pi^f [U(p_E^0, e_2^f) - U(P_E^f, e_2^f)] \right],$$

and therefore, we need to consider the contract to minimize  $B$ .

Proposition 1 provides the optimal complete contract.

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<sup>10</sup>Recall that a non-promoted worker receives payoff  $v(p_H)$ .

**Proposition 1.** *Let*

$$\xi(p_E^0) := Q(p_E^0)[U(P_E^f) - v(p_H)] + [U(p_E^0) - U(P_E^f)].$$

*The following  $(e_1, (e_2^o, \pi^o))$  constitutes the optimal contract:*

- (i) *(Strong complementarity,  $\Delta_L/\Delta_H < q_{LE}/q_{HE}$ .) The maximizers are promotion and continuation after success ( $\pi^s = 1, e_2^s = 1$ ), and worker retention after failure ( $\pi^f = 0$ , with the project run at full effort if and only if  $Q(P_E^f)\Pi > c$ ).*
- (ii) *(Strong substitutability,  $\Delta_L/\Delta_H > \frac{1-q_{LE}}{1-q_{HE}}$ .) The maximizers are continuation after success ( $\pi^s = 1, e_2^s = 1$ ). Promotion after failure is optimal if and only if  $\xi(p_E^0) < 0$ , with the project run at full effort if and only if  $(1 - Q(p_E^0))[Q(P_E^f)\Pi - c] > \min\{0, \xi(p_E^0)\}$ .*

Proposition 1 has two implications: one for the promotion rule, and one for the continuation decision after failure.

**Promotion rule: conventional under complementarity, inverted under substitutability.** Under complementarity (Proposition 1 (i)), the optimal rule conforms to conventional wisdom: the firm promotes after success ( $\pi^s = 1$ ) and withholds promotion after failure ( $\pi^f = 0$ ). Since  $U$  is increasing in  $p_E$ , the post-success belief  $P_E^s > p_E^0$  delivers a strictly higher reputation payoff  $U(P_E^s) > U(p_E^0)$  to the promoted worker, and the firm exploits this gap as the period-1 carrot.

Under substitutability (Proposition 1 (ii)), the rule inverts: *the firm promotes after failure* whenever  $\xi(p_E^0) < 0$ . The intuition rests on what failure reveals about the task. The posterior  $P_E^f < p_E^0$  implies that the project is perceived as harder, and under substitutability the period-2 reputational gap between success and failure is *wider* on harder projects (Lemma 1(vi)b): high-ability managers can still succeed where low-ability managers fail, so the outcome is more diagnostic of ability. The post-failure project thus carries higher reputational volatility than the unconditional prior project, and convex  $v$  assigns it strictly higher value:  $U(P_E^f) > U(p_E^0)$ . Promotion-after-failure is therefore not a punishment but a reward, and the firm uses it as the carrot whenever the gain  $U(P_E^f) - U(p_E^0)$  from handing the worker this higher-volatility lottery exceeds the rent  $Q(p_E^0)[U(P_E^f) - v(p_H)]$ , and the condition is summarized as  $\xi(p_E^0) < 0$ .

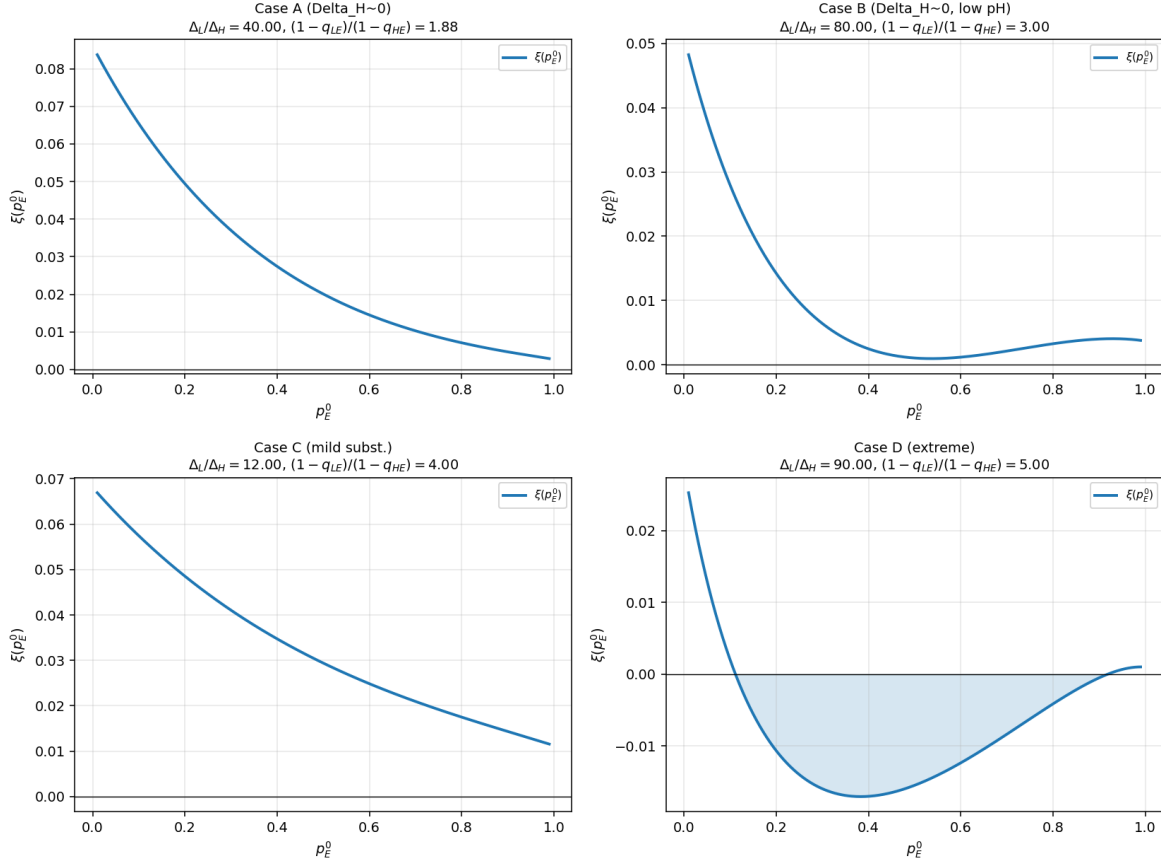


Figure 1: Plots of  $\xi(p_E^0)$  for  $v(x) = x^2$  under strong substitutability. Each panel uses different  $(q_{HE}, q_{HD}, q_{LE}, q_{LD}, p_H)$ . The solid (blue) curve shows that  $\xi(p_E^0)$  can take both positive and negative values; the negative region corresponds to the parameter ranges in which  $\pi^f = 1$  is optimal.

**Continuation at a loss.** The inverted promotion rule has a striking implication for project continuation. The post-failure reputational lottery pays off *only if* the firm actually runs the project in period 2: setting  $e_2^f = 0$  guarantees failure, collapses the lottery, and reduces the worker's payoff to  $v(p_H)$ . Hence, whenever it is optimal to promote after failure ( $\xi(p_E^0) < 0$ ), the firm must also commit to continuation ( $e_2^f = 1$ ), even when the project is expected to lose money ( $Q(P_E^f)\Pi < c$ ). Continuation is not a separate choice: it is the price the firm pays to preserve the reputational option value that motivates the worker. Formally,  $e_2^f = 1$  is optimal whenever  $(1 - Q(p_E^0))[Q(P_E^f)\Pi - c] > \xi(p_E^0)$ , which can hold under negative expected profit when  $\xi(p_E^0)$  is sufficiently negative.

Under complementarity, by contrast, the continuation decision decouples from the incentive problem. Because  $\pi^f = 0$ , the worker's continuation payoff is independent of  $e_2^f$ , and the firm chooses continuation purely on profitability grounds:  $e_2^f = 1$  if and only if  $Q(P_E^f)\Pi > c$ .

Note that the sign of  $\xi(p_E^0)$  depends on the parameters and on  $p_E^0$ . Figure 1 plots some numerical examples. The figure shows that  $\xi(p_E^0)$  can be negative on a substantial interior range of  $p_E^0$ , so that promotion after failure and the associated continuation at a loss can be optimal. The intuition is that the reputational bonus  $U(P_E^f) - U(p_E^0)$  from being assigned to the (perceived harder, higher-volatility) post-failure project must dominate the ex-post rent  $Q(p_E^0)[U(P_E^f) - v(p_H)]$  the firm pays the worker on the success branch. Near  $p_E^0 = 0$  or  $p_E^0 = 1$ , the reputational bonus vanishes (since  $P_E^f \rightarrow p_E^0$ ), so  $\xi$  is positive at the endpoints; the negative region lies in the interior.

| Case                    | Success   | Failure                        |
|-------------------------|-----------|--------------------------------|
| Strong complementarity  | Promotion | No promotion                   |
| Strong substitutability | Promotion | Promotion iff $\xi(p_E^0) < 0$ |

Table 1: Summary of Proposition 1 (Promotion decision)

In sum, complementarity yields the conventional rule of promoting after success, while substitutability inverts it: the firm promotes after failure and, to preserve the worker's reputational lottery, continues the project even at a loss (Table 1).

## 5 Discussion

### 5.1 No benching: sidelining as a punishment

In the baseline model, the firm can punish the worker not to promote. However, this arrangement may be costly in some cultures. For example, consider a seniority-based employment system coupled with strong job security, as in the classic Japanese model of lifetime employment. The firm can then neither keep an experienced employee indefinitely in a junior position nor dismiss them. As tenure accumulates, norms and implicit commitments push the employee upward in rank and pay, while employment protection rules out separation. Retaining the agent as a junior worker (the continuation that would deliver only  $v(p_H)$ ) is therefore unavailable. If the firm does not promote the agent onto the current project, it must still place them in a managerial role elsewhere, which yields  $U(p_E^0)$ .

This section considers this constraint: the firm should provide a managerial role to the worker in period 2. The choice is to assign the current engaging project, or an outside flesh project. If the worker declines the promotion, he goes outside and gets reputation utility  $v(p_H)$ .

To match this context, we refer to  $\pi^o = 1$  as *staying on* (the current project) and  $\pi^o = 0$  as *reassignment*. With this setting, the incentive compatibility condition for full effort in period 1 is

$$\beta_1 \geq \tilde{B} := \frac{1}{Q(p_E^0)} \left[ c - Q(p_E^0) \left[ \pi^s [U(p_E^s, e_2^s) - U(p_E^0)] - \pi^f [U(p_E^f, e_2^f) - U(p_E^0)] \right] \right. \\ \left. + \pi^f [U(p_E^0, e_2^f) - U(p_E^f, e_2^f)] \right],$$

The optimal contract is summarized as follows.

**Proposition 2.** *At the optimal contract, the following properties hold.*

(i) (Strong complementarity case,  $\Delta_L/\Delta_H < \frac{q_{LE}}{q_{HE}}$ ).

(a) The maximizers are staying on and continuation after success ( $\pi^s = 1, e_2^s = 1$ )

(b) After failure branch, reassignment and continuation ( $\pi^f = 0, e_2^f = 1$ ) is optimal if  $(1 - Q(p_E^0))[Q(p_E^f)\Pi - c] > Q(p_E^0)[U(p_E^0) - v(p_H)]$ , and otherwise, staying on but sidelining ( $\pi^f = 1, e_2^f = 0$ ) is optimal.

(ii) (Strong substitutability case,  $\Delta_L/\Delta_H > \frac{1-q_{LE}}{1-q_{HE}}$ ).

(a) The maximizers are reassignment and continuation after success ( $\pi^s = 0, e_2^s = 1$ )

(b) After failure branch, staying on and continuation ( $\pi^f = 1, e_2^f = 1$ ) is optimal if  $(1 - Q(p_E^0))[Q(p_E^f)\Pi - c] > \xi(p_E^0)$ ; otherwise, staying on and sidelining ( $\pi^f = 1, e_2^f = 0$ ) is optimal.

Proposition 2 has two main differences from Proposition 1. The first difference appears in the continuation decision after failure. Under complementarity, the firm may become willing to sideline a failed project (i.e.,  $e_2^f = 0$ ), even when continuation is expected to be profitable (i.e.,  $Q(p_E^f)\Pi > c$ )<sup>11</sup>, because sidelining serves as a disciplinary device: if the firm sets  $e_2^f = 0$ , the promoted worker earns only  $v(p_H)$  instead of  $U(p_E^0)$ , giving the firm a tool to deter shirking.

By contrast, under substitutability, the firm is more inclined to proceed with the project despite anticipated losses if  $\xi(p_E^0) > 0$ . The quantity  $\xi(p_E^0)$  captures the net reputational incentive effect of

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<sup>11</sup>Note that  $[U(p_E^0) - v(p_H)] > 0$ .



project continuation after failure. To capture the intuition, we rearrange  $\xi$  as follows.

$$\xi(p_E^0) = Q(p_E^0)[U(p_E^0) - v(p_H)] - (1 - Q(p_E^0))[U(P_E^f) - U(p_E^0)]$$

The first term,  $Q(p_E^0)[U(p_E^0) - v(p_H)]$ , is the *punishment value of sidelining*. As in the baseline model, the second term,  $(1 - Q(p_E^0))[U(P_E^f) - U(p_E^0)]$ , is the *reputational bonus* from managing the current project with an updated belief  $P_E^f$  rather than the prior  $p_E^0$ . When  $\xi(p_E^0) < 0$ , the reputational bonus from continuation dominates the punishment value of sidelining.

Another difference appears in the assignment rule after success in substitutability case: although  $\pi^s = 1$  in Proposition 1,  $\pi^s = 0$  in Proposition 2. Under substitutability, the current project becomes unattractive in terms of reputation gain. Committing to reassignment after success branch eliminates this concern for the worker.

Now Table 2 summarizes the assignment decision.

| Case                    | Success      | Failure     |
|-------------------------|--------------|-------------|
| Strong complementarity  | Stayning on  | Conditional |
| Strong substitutability | Reassignment | Stayning on |

Table 2: Summary of Proposition 2 (Assignment decision)

## 5.2 Imperfect sabotage and graded promotions

Our baseline model assumes that effort is binary, with  $e = 0$  leading to certain project failure. We now relax this by assuming that when  $e = 0$ , the project succeeds with probability  $\gamma Q$ , where  $\gamma > 0$ . We strengthen the baseline assumption  $Q(p_E^0)\Pi > c$  to its natural analog in this richer setting; at the prior, full effort dominates sidelining:

$$(1 - \gamma)Q(p_E^0)\Pi > c. \quad (3)$$

Under this richer setting, the firm has three options after failure: continue ( $e_2^f = 1$ ), promote and sideline ( $e_2^f = 0$  and  $\pi^f = 1$ ), or retain the worker as a junior ( $\pi^f = 0$ ). Because promotion after failure can be a prize that motivates first-period effort under substitutability, these options form a spectrum of *graded promotions*: sidelining is a genuine but partial reward, intermediate between

full continuation and no promotion. The firm calibrates the intensity of the reputational reward, and sidelining (i.e., promoting the worker while scaling back the project) emerges as the interior choice. Note that sidelining here is a prize, in contrast to Proposition 2, where it serves as a punishment.

Define  $U^\gamma(p_E, e) = U(p_E, (1 - \gamma)e + \gamma)$ , so that  $U^\gamma(p_E, 1) = U(p_E, 1)$  and  $U^\gamma(p_E, 0) = U(p_E, \gamma)$ , and let  $P_E^s = P_E^s(p_E^0, 1)$ ,  $P_E^f = P_E^f(p_E^0, 1)$ . As shown in the proof, the two outcome branches are separable: on each branch  $o \in \{s, f\}$ , the firm independently chooses among continuation ( $\pi^o = 1, e_2^o = 1$ ), sidelining ( $\pi^o = 1, e_2^o = 0$ ), and worker retention ( $\pi^o = 0$ ). For each branch, let

$$\begin{aligned} C^f(e) &:= (1 - \gamma Q(p_E^0))U^\gamma(p_E^0, e) - (1 - Q(p_E^0))U^\gamma(P_E^f, e) - (1 - \gamma)Q(p_E^0)v(p_H), \\ C^s(e) &:= Q(p_E^0)[\gamma U^\gamma(p_E^0, e) - U^\gamma(P_E^s, e) + (1 - \gamma)v(p_H)], \\ N_o^{\text{cont}} &:= (1 - Q(p_E^0))[Q(P_E^o)\Pi - c] - \frac{C^o(1)}{(1 - \gamma)Q(p_E^0)}, \\ N_o^{\text{side}} &:= (1 - Q(p_E^0))\gamma Q(P_E^o)\Pi - \frac{C^o(0)}{(1 - \gamma)Q(p_E^0)}, \\ N_o^{\text{ret}} &:= (1 - Q(p_E^0))\max\{Q(P_E^o)\Pi - c, \gamma Q(P_E^o)\Pi\}, \end{aligned}$$

Here  $N_o^\bullet$  ( $o \in \{s, f\}, \bullet \in \{\text{cont}, \text{side}, \text{ret}\}$ ) is the firm's payoff (period-2 expected profit net of the incentive payment  $\beta_1$ ) under each post- $o$  arrangement, up to a term common across the three.  $C^o$  is the current worker's reputation cost of promotion as the manager of the current project after the outcome is known. If  $C^o < 0$ , the promotion is a prize while it is cost when  $C^o > 0$ .

**Proposition 3.** *Fix  $\gamma \in (0, 1)$ . The two branches are separable, and the post- $o$  arrangement independently maximizes  $\{N_o^{\text{cont}}, N_o^{\text{side}}, N_o^{\text{ret}}\}$  for each  $o \in \{s, f\}$ . Moreover, under strong complementarity ( $\Delta_L/\Delta_H < q_{LE}/q_{HE}$ ), the maximizers are promotion and continuation after success ( $\pi^s = 1, e_2^s = 1$ ), and worker retention after failure ( $\pi^f = 0$ , with the project run at full effort if and only if  $(1 - \gamma)Q(P_E^f)\Pi > c$ ).*

Proposition 3 shows that sidelining is never optimal under strong complementarity. The firm's optimal post-failure arrangement is either continuation or worker retention, exactly as in the baseline.

Under substitutability, in contrast, all three arrangements (continuation, sidelining, and worker retention) can be optimal depending on the parameter configuration. Figure 2 plots the optimal promotion scheme after failure under substitutability. After failure the worker's reputation payoff is

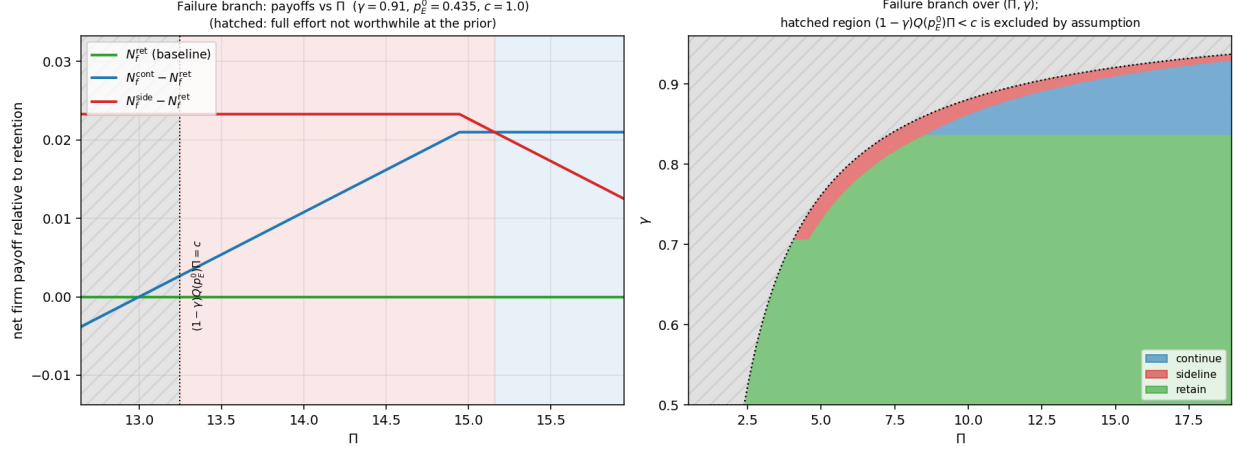


Figure 2: Failure-branch payoffs under strong substitutability and  $v(p) = p^2 - (p_H)^2$  ( $p_H = \frac{1}{2}$ ,  $(q_{HE}, q_{HD}, q_{LE}, q_{LD}) = (0.99, 0.93, 0.97, 0.53)$ ,  $p_E^0 = 0.435$ ,  $c = 1$ ). Left: the continuation and sidelining payoffs  $N_f^{\text{cont}}$ ,  $N_f^{\text{side}}$  measured against retention  $N_f^{\text{ret}}$  (the zero baseline), as functions of  $\Pi$  at  $\gamma = 0.91$ ; sidelining is optimal on the shaded interval inside the profit-loss band, and the hatched region violates (3). Right: the optimal post-failure arrangement over  $(\Pi, \gamma)$ ; the hatched region  $(1 - \gamma)Q(p_E^0)\Pi < c$  is excluded by (3).

ordered  $U(P_E^f, 1) > U(P_E^f, \gamma) > v(p_H)$  across continuation, sidelining, and retention, so sidelining is a *partial* reward, not a disguised punishment: a promoted-and-sidelined worker is strictly better off than one kept as a junior. Its primary role is to replace loss-making continuation.

As for the after-success branch, sidelining is never optimal under (3) in a profit sense. As success raises, the success probability ( $Q(P_E^s) > Q(P_E^0)$ ), so  $(1 - \gamma)Q(P_E^s)\Pi > c$  and the post-success project remains worth full effort: it is never sidelined. As this argument compares only profits, it leaves open whether the *incentive* channel might favor sidelining a successful worker (lowering the success-branch reward to curb the rent from shirking). Under several assumptions, it cannot after succes branch.

**Proposition 4** (No sidelining after success). *Suppose that  $v$  satisfies either one of the following conditions:*

1.  $v(p') = 0$  for each  $p' \leq p_H$
2.  $v''' \geq 0$  and  $\kappa(s, f) \geq 2 - (s - f) \frac{v'''(f)}{v''(f)}$  for each  $s, f \in (0, 1)$  with  $s > p_H > f$ . Here,  $\kappa(s, f) := \frac{v''(f)(s-f)^2}{v(s)-v(f)-(s-f)v'(f)}$ .

Consequently, for every  $\gamma \in (0, 1)$  and under (3), sidelining after success is never optimal.

The first condition assumes the option-value utility (1). The second condition is satisfied if, for example,  $v(x) = x^n$  for each  $n \geq 2$  and  $v(x) = \exp(ax)$  for each  $a > 0$ .

### 5.3 Retaining the Incumbent vs. Promoting a Successor

The basic model assumes that in the second period, the first-period manager exits, and a new manager is either internally promoted or externally hired. However, it is natural to consider an alternative: retaining the current manager.

This section examines whether the decision not to promote the current worker can be interpreted as a choice to retain the incumbent manager. Indeed, if the current project succeeds, it signals the high competence of the current manager and provides an incentive for the firm to retain her in the second period.

However, whether this retention is optimal depends on how such a decision affects the incentive structure for the current worker. From the insights of Proposition 1, the costs of retaining the incumbent manager are as follows. In the complementarity case ( $\Delta_L/\Delta_H < \frac{q_{LE}}{q_{HE}}$ ), as promoting the current worker incentivizes the current worker when the project succeeds, retaining the incumbent manager reduces the current worker's incentive for effort, which implies an increase in incentive pay. Therefore, there is a tradeoff for the firm. In contrast, in the substitutability case ( $\Delta_L/\Delta_H > \frac{1-q_{LE}}{1-q_{HE}}$ ), as not promoting the current worker is optimal when the project succeeds, there seems to be no tradeoff.

The above discussion ignores the cost of retaining the current manager. We need to consider the current manager's reputation gain, which specifies the payment to the current manager for the reappointment. Below, we develop a formal model for reappointing the current manager.

Other than the choice of the second-period manager, the setting is common to the basic model. As the first-period manager's outside option, we have the following possibilities. One is the benefit from the current reputation  $v(P_H^o)$ . An example situation is that another firm offers a contract to the incumbent manager for managing the outside project. In such a case, the incumbent manager has further reputation gains from managing the outside project, denoted by  $U$ , which will be calculated in the proof. We consider the reservation utility as the sum of them,  $v(P_H^o) + U$ . We assume that the outside project has the same prior about the ease,  $p_E^0$ .

The firm has three options: replace the manager with the current worker, employ an external worker, or reappoint the current manager. If the firm replaces the manager with the current worker or hires an external worker, the manager goes to another department and receives benefits from his reputation. The current and external workers' management ability is high with probability  $p_H$ . To retain the first-period manager, the firm makes an additional payment  $\mu \geq 0$ . To simplify, we assume that the firm can employ workers whose reputation is  $p_H$  with a cost of 0.

Further, assume that  $v$  is

$$v(p'_H) = \max\{(p'_H - p_H), 0\} \cdot \Pi \cdot [p_E^0 \cdot (q_{HE} - q_{LE}) + (1 - p_E^0) \cdot (q_{HD} - q_{LD})].$$

This is consistent with the assumption that the firm can employ workers with a reputation  $p_H$  at cost 0. Let  $R_{o_1}$  be the difference in the second-period benefits between the reappointment of the current manager and employing a worker with reputation  $p_H$  when the first-period outcome is  $o_1 \in \{s, f\}$ . That is,

$$R_{o_1} = \underbrace{[Q_{o_1}\Pi - c] - \mu_{o_1}}_{\text{profit of retain}} - \underbrace{[Q(P_E^{o_1})\Pi - c]}_{\text{profit of replace}},$$

where  $Q_{o_1} = P_H^{o_1}(P_E^{o_1}q_{HE} + (1 - P_E^{o_1})q_{HD}) + (1 - P_H^{o_1})(P_E^{o_1}q_{LE} + (1 - P_E^{o_1})q_{LD})$  is the success probability of the current project with the reappointed manager.

**Proposition 5.** *Suppose that  $q_{LD} > 1/2$ . Then, if  $\Delta_L/\Delta_H < \frac{q_{LE}}{q_{HE}}$ ,  $R_s > 0 > R_f$ , and if  $\Delta_L/\Delta_H > \frac{1-q_{LE}}{1-q_{HE}}$ ,  $R_s, R_f < 0$ .*

This shows that in the complementarity case, since reappointing the current manager yields a benefit, there is a trade-off between leveraging this benefit and incentivizing the first-period worker (Proposition 1 (i)). This trade-off may lead the firm to forgo reappointment despite its direct advantage to maintain strong incentives for the current manager. Furthermore, in this case, the firm does not hire externally.

In contrast, if  $o_1 = f$ , the benefit of reappointing the current manager is negative ( $R_f < 0$ ). Therefore, when the firm does not promote the current worker, it hires an external candidate instead, and the implementation condition for  $e_2^f$  remains unchanged.

In the substitutability case, there is no benefit from reappointing the current manager. Thus, according to Proposition 1 (ii), the optimal promotion rule remains unchanged even when reappointment is possible.

In conclusion, reappointment is beneficial only when the first-period project succeeds and when project ease and managerial ability are complementary.

## 5.4 Further Modeling Assumptions

### 5.4.1 Unobservable Efforts

Proposition 1 (ii) demonstrates that it is optimal to promote the current worker involved in a project when that project fails. This conclusion, however, relies on the assumption that effort is observable. If effort is unobservable, the *ratchet effect problem* (e.g., Freixas, Guesnerie, and Tirole, 1985; Ishida, 2012) arises: unobserved shirking can lead to the current project failing and appearing more difficult.<sup>12</sup> This apparent difficulty can enhance the next manager's reputation for undertaking a seemingly challenging project.

To illustrate this, we assume that effort is partially observable: a worker's shirking ( $e_1 = 0$ ) is observed by the firm with probability  $\rho \in (0, 1)$ . However, this observation is unverifiable, preventing the firm from differentiating its treatment of the worker based on whether shirking was detected.

Under this assumption, the incentive-compatibility condition for the period-1 worker to exert effort is the following.

$$\beta_1 \geq B^\rho := \frac{1}{Q(p_E^0)} \left[ c - Q(p_E^0) \left[ \pi^s [U(P_E^s, e_2^s) - v(p_H)] - \pi^f [U(P_E^f, e_2^f) - v(p_H)] \right] \right. \\ \left. + \rho \pi^f [U(p_E^0, e_2^f) - U(P_E^f, e_2^f)] \right].$$

The effect of unobservability appears when the project fails. Hence, the difference arises in  $\pi^f$ . The following Proposition shows that unobservability changes the optimal contract only on the failure branch, and only under substitutability.

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<sup>12</sup>In a companion paper, Yamagata and Tajika (2023), we mainly focus on the ratchet effects, and consider multiperiod dynamics of contracts with internal promotion.

**Proposition 6.** *Let*

$$\xi^\rho(p_E^0) := Q(p_E^0)[U(P_E^f) - v(p_H)] + \rho[U(p_E^0) - U(P_E^f)].$$

*The following  $(e_1, (e_2^o, \pi^o))$  constitutes the optimal contract:*

- (i) *(Strong complementarity,  $\Delta_L/\Delta_H < q_{LE}/q_{HE}$ .) The optimal contract coincides with that in Proposition 1(i).*
- (ii) *(Strong substitutability,  $\Delta_L/\Delta_H > \frac{1-q_{LE}}{1-q_{HE}}$ .) The maximizers are continuation after success ( $\pi^s = 1, e_2^s = 1$ ). Promotion after failure is optimal if and only if  $\xi^\rho(p_E^0) < 0$ , with the project run at full effort if and only if  $(1 - Q(p_E^0))[Q(P_E^f)\Pi - c] > \min\{0, \xi^\rho(p_E^0)\}$ .*

The intuition is as follows. Under complementarity, a more difficult project lowers the promoted manager's reputation utility (Lemma 2). Therefore, the firm never promote the current worker after failure, which implies that the ratchet effect is absent.

The opposite intuition applies to the substitutability case. Under substitutability, a more difficult project raises the promoted manager's reputation utility, which tempts the firm to promote the worker after failure. The worker then benefits from shirking: the project comes to look difficult, and the resulting promotion yields higher reputation utility. This is precisely the ratchet effect, which discourages the firm from promoting after failure.

These results reveal that how effort observability affects promotion decision depends on the production technology.

#### 5.4.2 Intermediate Substitutability

Most of our discussion has focused on two extreme cases:  $\Delta_L/\Delta_H$  is either sufficiently high or sufficiently low. This section examines whether our results can be extended to cases of intermediate substitutability. Proposition 7 provides a generalization of Lemma 2.

**Proposition 7.** *Suppose that  $v''' \geq 0$ . Let  $Q_a$  denote the success probability when the manager's ability is  $a \in \{L, H\}$ , and the probability that the project is easy is  $p_E$ . Then,  $U'(p_E) < 0$  if and*

only if

$$\frac{\Delta_L}{\Delta_H} > \frac{Q(p_E) + (1 - 2Q(p_E))Q_L}{Q(p_E) + (1 - 2Q(p_E))Q_H}. \quad (4)$$

Although we can specify the threshold explicitly, it depends on the probability that the project is easy. However, as long as  $P_E^s$  and  $P_E^f$  remain on the same side of the threshold, our main results continue to hold even under intermediate substitutability.<sup>13</sup>

### 5.4.3 The Role of Commitment

Our model assumes that the firm commits to the promotion rule and the effort profile  $(e_2^s, e_2^f)$  before the first-period outcome is realized. This assumption is standard when analyzing optimal contracts, but it is worth examining where commitment is binding.

Under complementarity, Proposition 1 (i-c) shows that the firm's decision on project continuation is standard (it continues if and only if  $Q(P_E^f)\Pi > c$ ). Therefore, the value of commitment is absent. Under substitutability, Proposition 1 (ii-c) shows that the firm may continue a loss-making project ( $Q(P_E^f)\Pi < c$ ) to preserve the worker's reputational option value. Without commitment, the firm would sideline the project, undermining the reputational bonus that incentivizes first-period effort.

## 6 Conclusion

We develop a two-period model of optimal promotion design under dual uncertainty about managerial ability and task ease. Our central result is that the optimal promotion rule depends on the modularity of the production technology. Under complementarity with convex reputation utility, promotion should reward success; under substitutability, promotion optimally follows failure. When effort is only partially observable, a ratchet effect modifies but does not overturn these insights.

Two broader implications follow. First, promotion design must account for how outcomes are interpreted, not just whether they are positive or negative. Rewarding success can backfire when success is attributed to easy tasks, and recognizing failure can signal resilience under difficulty.

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<sup>13</sup>We randomly sampled 10,000 parameter configurations  $(q_{HE}, q_{HD}, q_{LE}, q_{LD}, p_H)$  uniformly subject to the ordering restrictions. Condition (4) was either satisfied or violated for all  $p_E \in [0, 1]$  in more than 80% of configurations. See the replication code at <https://github.com/Ttajika/Yamagata-Tajika-Promotion-TaskEase>



Second, promotion policy interacts with the continuation decision: under substitutability, maintaining the reputational option value can rationalize continuing projects with negative expected profits.

Future work could study interacting projects or longer horizons, where task ease evolves dynamically (e.g., as a Markov process), to further examine the interplay between learning, incentives, and assignment.

## A Proofs

*Proof of Lemma 1.* (i)  $P_E^s$  and  $P_E^f$  are calculated as

$$P_E^s(p_E, e) = \frac{p_E E[q_{aE}]e}{p_E E[q_{aE}]e + (1 - p_E)E[q_{aD}]e} = \frac{p_E E[q_{aE}]}{p_E E[q_{aE}] + (1 - p_E)E[q_{aD}]},$$

$$P_E^f(p_E, e) = \frac{p_E(1 - E[q_{aE}]e)}{p_E(1 - E[q_{aE}]e) + (1 - p_E)(1 - E[q_{aD}]e)},$$

where  $E[q_{ad}] = p_H q_{Hd} + (1 - p_H)q_{Ld}$  for each  $d \in \{E, D\}$ . This shows that  $P_E^s$  is independent of  $e$ . Further, by the differentiation and the assumption that  $q_{aE} > q_{aD}$ ,  $P_E^f$  is decreasing in  $e$ .

(ii) Calculating  $P_E^s(p_E, e) - p_E$  and  $P_E^f(p_E, e) - p_E$  yields that

$$P_E^s(p_E, e) - p_E = \frac{p_E(1 - p_E)[E[q_{aE}] - E[q_{aD}]]}{p_E E[q_{aE}] + (1 - p_E)E[q_{aD}]} > 0,$$

$$P_E^f(p_E, e) - p_E = \frac{p_E(1 - p_E)(E[q_{aD}] - E[q_{aE}])e}{p_E(E[q_{aD}] - E[q_{aE}])e + (1 - E[q_{aD}]e)} \leq 0.$$

The inequality follows from the assumption that  $q_{aE} > q_{aD}$ .

(iii) By differentiation,

$$\frac{\partial P_E^s(p_E, e)}{\partial p_E} = \frac{E[q_{aE}]E[q_{aD}]}{(p_E E[q_{aE}] + (1 - p_E)E[q_{aD}])^2} > 0$$

$$\frac{\partial P_E^f(p_E, e)}{\partial p_E} = \frac{(1 - E[q_{aE}]e)(1 - E[q_{aD}]e)}{[p_E(1 - E[q_{aE}]e) + (1 - p_E)(1 - E[q_{aD}]e)]^2} > 0.$$

Further, as  $E[q_{aE}] > E[q_{aD}]$ ,  $\frac{\partial^2 P_E^s(p_E, e)}{\partial (p_E)^2} < 0$  and  $\frac{\partial^2 P_E^f(p_E, e)}{\partial (p_E)^2} > 0$ .

(iv) Note that  $P_H^s$  and  $P_H^f$  are calculated as follows.

$$P_H^s(p_E, e) = \frac{p_H(p_E q_{HE} + (1 - p_E)q_{HD})}{p_H(p_E q_{HE} + (1 - p_E)q_{HD}) + (1 - p_H)(p_E q_{LE} + (1 - p_E)q_{LD})},$$

$$P_H^f(p_E, e) = \frac{e p_H(p_E(1 - q_{HE}) + (1 - p_E)(1 - q_{HD})) + (1 - e)p_H}{1 - Q(p_E)e},$$

Note that  $1 - Q(p_E)e = e[p_H(p_E(1 - q_{HE}) + (1 - p_E)(1 - q_{HD})) + p_L(p_E(1 - q_{LE}) + (1 - p_E)(1 - q_{LD}))] + (1 - e)$ .

Then, (iv) is demonstrated by the following calculations:

$$P_H^s(p_E, e) - p_H = \frac{p_H(1 - p_H)[p_E(q_{HE} - q_{LE}) + (1 - p_E)(q_{HD} - q_{LD})]}{p_H(p_E q_{HE} + (1 - p_E)q_{HD}) + p_L(p_E q_{LE} + (1 - p_E)q_{LD})} > 0,$$

$$p_H - P_H^f(p_E, e) = \frac{e p_H(1 - p_H)[p_E(q_{HE} - q_{LE}) + (1 - p_E)(q_{HD} - q_{LD})]}{1 - Q(p_E)e} \geq 0.$$

(v) The expectation is calculated as follows.

$$E[P_H^o(p_E, e)] = Q(p_E)eP_H^s(p_E, e) + (1 - Q(p_E)e)P_H^f(p_E, e) = p_H.$$

This completes the proof.

(vi) Differentiating  $P_H$  with respect to  $p_E$  yields

$$\frac{\partial P_H^s}{\partial p_E}(p_E) = \frac{p_H p_L(Q_L \Delta_H - Q_H \Delta_L)}{Q^2},$$

$$\frac{\partial P_H^f}{\partial p_E}(p_E) = e p_H p_L \frac{-[(1 - Q_L e)\Delta_H - (1 - Q_H e)\Delta_L]}{(1 - Qe)^2}.$$

where  $Q_H = p_E q_{HE} + (1 - p_E)q_{HD}$ ,  $Q_L = p_E q_{LE} + (1 - p_E)q_{LD}$ ,

$$Q = p_H Q_H + (1 - p_H)Q_L, \quad \Delta_H = q_{HE} - q_{HD}, \quad \Delta_L = q_{LE} - q_{LD}.$$

Note that when  $e = 0$ ,  $\frac{\partial P_H^f}{\partial p_E}(p_E) = 0$ . So, focus on the case where  $e = 1$ . Then,

$$\frac{\partial P_H^s}{\partial p_E}(p_E) > 0 \iff \frac{Q_L}{Q_H} > \frac{\Delta_L}{\Delta_H} \quad (5)$$

$$\frac{\partial P_H^f}{\partial p_E}(p_E) > 0 \iff \frac{\Delta_L}{\Delta_H} > \frac{1 - Q_L}{1 - Q_H} \quad (6)$$

Now, suppose that  $\Delta_L/\Delta_H < 1$ . Then, inequality (6) is violated and inequality (5) holds if

$$1 > \min_{p_E} \frac{Q_L}{Q_H} = \frac{q_{LE}}{q_{HE}} > \frac{\Delta_L}{\Delta_H}.$$

This shows that  $P_H^s(p_E)$  is increasing and  $P_H^f(p_E)$  is decreasing for low  $\Delta_L/\Delta_H$ .

Next, suppose that  $\Delta_L/\Delta_H > 1$ . Then inequality (5) is violated, and inequality (6) holds if

$$\frac{\Delta_L}{\Delta_H} > \max_{p_E} \frac{1 - Q_L}{1 - Q_H} = \frac{1 - q_{LE}}{1 - q_{HE}} > 1.$$

This shows that  $P_H^s(p_E)$  is decreasing and  $P_H^f(p_E)$  is increasing for high  $\Delta_L/\Delta_H$ .

□

*Proof of Lemma 2.* (i) Immediate from Lemma 1 (iv) and the convexity of  $v$

(ii) From Lemma 1 (vi), if  $\Delta_L/\Delta_H < q_{LE}/q_{HE}$ ,  $P_H^s(p_E, 1)$  increases and  $P_H^f(p_E, 1)$  decreases as  $p_E$  increases while  $E_o[P_H^o(p_E, 1)] = p_H$  for each  $p_E$  (from Lemma 1 (v)). Recall that  $P_H^s(p_E, 1) > p_H > P_H^f(p_E, 1)$ . Then, by the convexity of  $v$  (which is not linear around  $p_H$ ),  $U(p_E, 1)$  increases from the following well-known fact.

**Fact 1.** Suppose that  $a > a' > b' > b$ , and  $pa + (1 - p)b = p'a' + (1 - p')b'$ . Then,  $pv(a) + (1 - p)v(b) \geq p'v(a') + (1 - p')v(b')$  with strict inequality if  $v$  is not affine on  $[b, a]$ .

This fact is obvious as  $(a, b; p)$  is a mean-preserving spread of  $(a', b'; p')$ .

(iii) The proof is parallel to case (ii).

□

*Proof of Proposition 1.* Recall that

$$B = \frac{1}{Q(p_E^0)} \left[ c - Q(p_E^0) \left[ \pi^s [U(P_E^s, e_2^s) - v(p_H)] - \pi^f [U(P_E^f, e_2^f) - v(p_H)] \right] \right. \\ \left. + \pi^f [U(p_E^0, e_2^f) - U(P_E^f, e_2^f)] \right]$$

- (i) Suppose that  $\Delta_L/\Delta_H < \frac{q_{LE}}{q_{HE}}$ . From Lemma 2 (ii),  $U$  is increasing in  $p_E$ . Then, as  $P_E^s > p_E^0$ ,  $U(P_E^s, 1) > U(P_E^s, 0) = v(p_H)$ , which implies that  $\pi^s = 1$  is optimal if  $e_2^s = 1$  and  $\pi^s = 0$  is optimal if  $e_2^s = 0$ . Also,

$$Q(p_E^0) \frac{\partial B}{\partial \pi^f} = Q(p_E^0) \underbrace{[U(P_E^f, e_2^f) - v(p_H)]}_{\text{positive}} + \underbrace{U(p_E^0, e_2^f) - U(P_E^f, e_2^f)}_{\text{positive}}.$$

Then, to minimize  $B$ ,  $\pi^f = 0$  is optimal. Note that as the project ends in period 2, we need not consider the effect of reputation in the incentive pay  $(\beta_2^s, \beta_2^f)$ . Therefore, the optimal payment to incentivize the effort of the second-period worker is  $\beta_2^o = c/Q(P_E^o)$  for each  $o \in \{s, f\}$ .

By using the relation  $\pi^s = e_2^s$  and  $\pi^f = 0$ , the total profit is

$$\left[ Q(p_E^0) [Q(P_E^s) \Pi - c] e_2^s + (1 - Q(p_E^0)) [Q(P_E^f) \Pi - c] e_2^f \right] \\ + Q(p_E^0) [\Pi + [U(P_E^s) - U(p_E^0)] e_2^s] - c.$$

Then,  $e_2^s = 1$  is optimal as  $Q(P_E^s) \Pi - c > Q(p_E^0) \Pi - c > 0$ .  $e_2^f = 1$  if and only if  $(1 - Q(p_E^0)) [Q(P_E^f) \Pi - c] > 0$ .

Finally, we check the optimality to enforce the first period worker to effort. If  $e_1 = 0$ . Then, the total profit is  $[Q(p_E^0) \Pi - c]$ , which yields lower profit than  $e_1 = 1$ .

- (ii) Suppose that  $\Delta_L/\Delta_H > \frac{1-q_{LE}}{1-q_{HE}}$ . Then, as  $U(P_E^s, e_2^s) > v(p_H)$ ,  $\pi^s = 1$  is optimal. Also,

$$Q(p_E^0) \frac{\partial B}{\partial \pi^f} = Q(p_E^0) \underbrace{[U(P_E^f, e_2^f) - v(p_H)]}_{\text{positive}} + \underbrace{U(p_E^0, e_2^f) - U(P_E^f, e_2^f)}_{\text{negative}}$$

Then, if  $e_2^f = 1$ ,  $Q(p_E^0) \frac{\partial B}{\partial \pi^f} = \xi(p_E^0) := Q(p_E^0)[U(P_E^f) - v(p_H)] + [U(p_E^0) - U(P_E^f)]$ .  
Therefore,  $\pi^f = 1$  if and only if  $\xi(p_E^0) < 0$ . On the contrary, if  $e_2^f = 0$ ,  $Q(p_E^0) \frac{\partial B}{\partial \pi^f} = 0$ .  
Therefore, any  $\pi^f \in [0, 1]$  is optimal. Now the total profit when  $e_1 = 1$  is enforced is

$$\begin{aligned} & \left[ Q(p_E^0)[Q(P_E^s)\Pi - c]e_2^s + (1 - Q(p_E^0))[Q(P_E^f)\Pi - c]e_2^f \right] \\ & + Q(p_E^0)\Pi - c + Q(p_E^0)[U(P_E^s, e_2^s) - v(p_H)] - \xi(p_E^0)e_2^f\pi^f \end{aligned}$$

Then,  $e_2^s = 1$  is optimal.  $e_2^f = 1$  is optimal if

$$(1 - Q(p_E^0))[Q(P_E^f)\Pi - c] > \xi(p_E^0)\pi^f.$$

As  $\pi^f = 1$  is optimal if and only if  $\xi(p_E^0) < 0$ ,  $\xi(p_E^0)\pi^f = \min\{0, \xi(p_E^0)\}$ .

Finally,  $e_1 = 1$  is optimal as the total profit exceeds  $Q(p_E^0)\Pi - c$ .

□

*Proof of Proposition 3.* We use the following claim, whose proof is given below.

**Claim 1.** Fix  $\gamma \in (0, 1)$ .

1. If  $\Delta_L/\Delta_H < q_{LE}/q_{HE}$  (strong complementarity), then  $\frac{\partial^2 U(p_E, \gamma)}{\partial p_E \partial \gamma} > 0$ .
2. If  $\Delta_L/\Delta_H > (1 - q_{LE})/(1 - q_{HE})$  (strong substitutability), then  $\frac{\partial^2 U(p_E, \gamma)}{\partial p_E \partial \gamma} < 0$ .

Using  $U^\gamma$ , the incentive-compatibility condition is

$$\begin{aligned} & Q(p_E^0)[\beta_1 + \pi^s U^\gamma(P_E^s, e_2^s) + (1 - \pi^s)v(p_H)] + (1 - Q(p_E^0))[\pi^f U^\gamma(P_E^f, e_2^f) + (1 - \pi^f)v(p_H)] - c \geq \\ & \gamma Q(p_E^0)[\beta_1 + \pi^s U^\gamma(p_E^0, e_2^s) + (1 - \pi^s)v(p_H)] + (1 - \gamma Q(p_E^0))[\pi^f U^\gamma(p_E^0, e_2^f) + (1 - \pi^f)v(p_H)]. \end{aligned} \tag{7}$$

(7) is rearranged as follows.

$$\beta_1 \geq B^\gamma := \frac{1}{(1 - \gamma)Q(p_E^0)} \left[ c + \pi^s C^s(e_2^s) + \pi^f C^f(e_2^f) \right].$$

The firm maximizes total profit, i.e., the period-2 expected profit minus  $\beta_1$ , where  $\beta_1$  binds at  $B^\gamma$ , the right-hand side of the displayed inequality divided by  $(1 - \gamma)Q(p_E^0)$ . Both that right-hand side and the period-2 expected profit are additively separable across the two outcome branches (the  $\pi^s$ - and  $\pi^f$ -terms above, and the success and failure profits), so the firm optimizes the success and failure branches independently. On branch  $o$ , the three arrangements yield the net payoffs  $N_o^{\text{cont}}$ ,  $N_o^{\text{side}}$ ,  $N_o^{\text{ret}}$  defined above, and the firm chooses their maximizer. We now determine the maximizers in each case.

- (i) Suppose that  $\Delta_L/\Delta_H < \frac{q_{LE}}{q_{HE}}$ . In this case, we have that  $U^\gamma(P_E^s, e_2^s) > U^\gamma(p_E^0, e_2^s) > v(p_H)$  for each  $e_2^s$ . Therefore,  $C^s(e) < 0$  and thus  $\pi^s = 1$  is optimal. Furthermore,  $C^s(1) < C^s(0)$  if and only if

$$U^\gamma(P_E^s, 1) - U^\gamma(P_E^s, 0) > \gamma[U^\gamma(p_E^0, 1) - U^\gamma(p_E^0, 0)].$$

By Claim 1,  $\frac{\partial^2 U(p_E, \gamma)}{\partial p_E \partial \gamma} > 0$ , which implies that  $U^\gamma(P_E^s, 1) - U^\gamma(P_E^s, 0) > [U^\gamma(p_E^0, 1) - U^\gamma(p_E^0, 0)] > \gamma[U^\gamma(p_E^0, 1) - U^\gamma(p_E^0, 0)]$ .

Moreover, by (3) and  $Q(P_E^s) > Q(p_E^0)$ , we have  $(1 - \gamma)Q(P_E^s)\Pi > c$ , so full effort also maximizes the success-branch profit. Both incentive and profit favor  $e_2^s = 1$ ; continuation after success thus dominates sideling after success, and  $e_2^s = 1$  is optimal.

As for  $\pi^f$ , note that  $U^\gamma(P_E^s, e_2^s) \geq U^\gamma(p_E^0, e_2^s) > v(p_H)$ .

$$\begin{aligned} C^f(e_2^f) &= (1 - \gamma Q(p_E^0))U^\gamma(p_E^0, e_2^f) - (1 - Q(p_E^0))U^\gamma(P_E^f, e_2^f) + (1 - \gamma)Q(p_E^0)v(p_H) \\ &\geq (1 - \gamma)Q(p_E^0)[U^\gamma(p_E^0, e_2^f) - v(p_H)] > 0. \end{aligned}$$

Therefore,  $\pi^f = 0$  is optimal: that is,  $C^f(e) > 0$  for each  $e$ , so promoting after failure only raises  $\beta_1$ . Since  $\pi^f = 0$ , the worker's payoff after failure equals  $v(p_H)$  regardless of  $e_2^f$ , so the project effort is a pure profitability choice: the project is run at full effort ( $e_2^f = 1$ ) if and only if  $Q(P_E^f)\Pi - c > \gamma Q(P_E^f)\Pi$ , i.e.,  $(1 - \gamma)Q(P_E^f)\Pi > c$ . Sideling ( $\pi^f = 1$ ,  $e_2^f = 0$ ) requires promoting after failure and is therefore dominated by worker retention; it is never optimal.

□

*Proof of Proposition 4.* To simplify, we normalize  $v$  so that  $v(p_H) = 0$ .

As  $(1 - \gamma)Q(P_E^s)\Pi - c > (1 - \gamma)Q(p_E^0)\Pi - c > 0$ ,  $N_s^{\text{cont}} < N_s^{\text{side}}$  only if  $C^s(1) > C^s(0)$ , equivalently,

$$\gamma > \frac{U^\gamma(P_E^s, 1) - U^\gamma(P_E^s, 0)}{U^\gamma(p_E^0, 1) - U^\gamma(p_E^0, 0)}. \quad (8)$$

Similarly,  $N_s^{\text{ret}} < N_s^{\text{side}}$  only if  $C^s(0) < 0$ , equivalently,

$$\gamma[U^\gamma(p_E^0, 0) - v(p_H)] < U^\gamma(P_E^s, 0) - v(p_H) \iff \gamma < \frac{U^\gamma(P_E^s, 0)}{U^\gamma(p_E^0, 0)}. \quad (9)$$

If inequalities (8) and (9) are satisfied,

$$\frac{U^\gamma(P_E^s, 1)}{U^\gamma(p_E^0, 1)} < \frac{U^\gamma(P_E^s, 0)}{U^\gamma(p_E^0, 0)}. \quad (10)$$

We examine the two cases:

1. Suppose that  $v(p) = 0$  for each  $p \leq p_H$ . Then, as  $P_H^f(p_E, e) < p_H$ ,  $\frac{U^\gamma(P_E^s, e)}{U^\gamma(p_E^0, e)}$  is constant with respect to  $e$ . Therefore, (10) never holds.
2. Suppose that  $v''' > 0$  and  $\kappa(s, f) \geq 2 - (s - f)\frac{v'''(f)}{v''(f)}$ .

We will show that  $\frac{U^\gamma(P_E^s, e)}{U^\gamma(p_E^0, e)}$  is increasing in  $e$ . To this end, as  $\frac{\partial U^\gamma(p, e)}{\partial e} > 0$ , it is sufficient to show that  $\frac{\frac{\partial U^\gamma(P_E^s, e)}{\partial e}}{\frac{\partial U^\gamma(p_E^0, e)}{\partial e}}$  is increasing in  $e$ . As  $P_E^s > p_E^0$ , This condition is equivalent to  $\frac{\partial}{\partial e} \ln \frac{\partial U^\gamma(p, e)}{\partial e}$  is increasing in  $p$ . By a calculation, we have that

$$\frac{\partial}{\partial e} \ln \frac{\partial U^\gamma(p, e)}{\partial e} = \frac{(1 - \gamma)Q(p)}{1 - Q(p)(e')} \kappa(P_H^s(p, e'), P_H^f(p, e')),$$

where  $e' = e + (1 - e)\gamma$ . Note that  $Q(p)/(1 - Q(p)e')$  is increasing in  $p$ . Furthermore, by strong substitutability,  $P_H^s(p, e')$  is decreasing in  $p$  and  $P_H^f(p, e')$  is increasing in  $p$ .

Therefore, if  $\frac{\partial \kappa(s, f)}{\partial s} < 0 < \frac{\partial \kappa(s, f)}{\partial f}$ ,  $\frac{\partial}{\partial e} \ln \frac{\partial U^\gamma(p, e)}{\partial e}$  is increasing in  $p$ .

Indeed,

$$\begin{aligned}\frac{\partial \kappa(s, f)}{\partial s} \leq 0 &\iff \frac{v(s) - v(f)}{s - f} \leq \frac{v'(s) + v'(f)}{2} \\ \frac{\partial \kappa(s, f)}{\partial f} \geq 0 &\iff \kappa(s, f) \geq 2 - (s - f) \frac{v'''(f)}{v''(f)}.\end{aligned}$$

The latter inequality is satisfied by the assumption. The former inequality is implied by the assumption  $v''' \geq 0$  and  $v'' \geq 0$ .

In summary, we have shown that  $\frac{U^\gamma(P_E^s, e)}{U^\gamma(P_E^0, e)}$  is nondecreasing in  $e$  and therefore, (10) never holds.

In either case, as (10) never holds,  $\max\{N_s^{\text{ret}}, N_s^{\text{cont}}\} \geq N_s^{\text{side}}$ . □

*Proof of Claim 1.* We first note that

$$\begin{aligned}\frac{\partial P_H^f(p_E, \gamma)}{\partial \gamma} &= \frac{p_H(1 - p_H)[Q_L - Q_H]}{(1 - Q(p_E)\gamma)^2} < 0. \\ \frac{\partial^2 P_H^f(p_E, \gamma)}{\partial \gamma \partial p_E} &= p_H(1 - p_H) \frac{(\Delta_L - \Delta_H)(1 - Q(p_E)\gamma) + 2\gamma Q'(p_E)(Q_L - Q_H)}{(1 - Q(p_E)\gamma)^3},\end{aligned}$$

where  $Q'(p_E) = p_H\Delta_H + (1 - p_H)\Delta_L$  and  $Q_L - Q_H = (\Delta_L - \Delta_H)p_E + (q_{LD} - q_{HD})$ .

Then, the cross derivative of  $U$  is calculated as follows.

$$\frac{\partial^2 U(p_E, \gamma)}{\partial p_E \partial \gamma} = \frac{1}{\gamma} \frac{\partial U(p_E, \gamma)}{\partial p_E} + (1 - Q(p_E)\gamma)v''(P_H^f(p_E, \gamma)) \frac{\partial P_H^f(p_E, \gamma)}{\partial p_E} \frac{\partial P_H^f(p_E, \gamma)}{\partial \gamma} + R,$$

where

$$\begin{aligned}R &= -\frac{1}{\gamma} v'(P_H^f(p_E, \gamma)) \frac{\partial P_H^f(p_E, \gamma)}{\partial p_E} - Q'(p_E)\gamma v'(P_H^f(p_E, \gamma)) \frac{\partial P_H^f(p_E, \gamma)}{\partial \gamma} \\ &\quad + (1 - Q(p_E)\gamma)v'(P_H^f(p_E, \gamma)) \frac{\partial^2 P_H^f(p_E, \gamma)}{\partial p_E \partial \gamma} \\ &= \frac{v'_f}{(1 - Q\gamma)^2} p_H(1 - p_H) \left[ -Q\gamma(\Delta_L - \Delta_H) + \gamma Q'(Q_L - Q_H) + (\Delta_L Q_H - \Delta_H Q_L)\gamma \right] = 0,\end{aligned}$$

where the bracket vanishes identically: substituting  $Q = p_H Q_H + (1 - p_H)Q_L$  and  $Q' = p_H\Delta_H + (1 - p_H)\Delta_L$ , the coefficient of each of  $Q_H\Delta_L$ ,  $Q_H\Delta_H$ ,  $Q_L\Delta_L$ ,  $Q_L\Delta_H$  is zero.



Since  $Q_H \geq Q_L$ , the first display gives  $\frac{\partial P_H^f(p_E, \gamma)}{\partial \gamma} < 0$ , and  $R = 0$  reduces the cross derivative to

$$\frac{\partial^2 U(p_E, \gamma)}{\partial p_E \partial \gamma} = \frac{1}{\gamma} \frac{\partial U(p_E, \gamma)}{\partial p_E} + (1 - Q(p_E) \gamma) v''(P_H^f) \frac{\partial P_H^f(p_E, \gamma)}{\partial p_E} \frac{\partial P_H^f(p_E, \gamma)}{\partial \gamma}. \quad (11)$$

We sign the two terms. By Lemma 1 (vi) with effective effort  $\gamma$ ,

$$\frac{\partial P_H^f(p_E, \gamma)}{\partial p_E} = \gamma p_H (1 - p_H) \frac{-[(1 - Q_L \gamma) \Delta_H - (1 - Q_H \gamma) \Delta_L]}{(1 - Q(p_E) \gamma)^2},$$

whose sign equals that of  $\Delta_L / \Delta_H - \frac{1 - Q_L \gamma}{1 - Q_H \gamma}$ . Because  $Q_H \geq Q_L$ , the threshold satisfies  $1 \leq \frac{1 - Q_L \gamma}{1 - Q_H \gamma} \leq \frac{1 - q_{LE}}{1 - q_{HE}}$  for every  $p_E \in (0, 1)$  and  $\gamma \in (0, 1)$  (it is increasing in  $\gamma$ , and its maximum over  $p_E$  is attained at  $\gamma = 1$ ). Moreover, since  $P_H^s$  is increasing and  $P_H^f$  decreasing in  $p_E$  (resp. the reverse) for every effective effort  $\gamma \in (0, 1)$  under strong complementarity (resp. substitutability), and  $E_o[P_H^o(p_E, \gamma)] = p_H$ , the mean-preserving-spread argument of Lemma 2 applies verbatim to  $U(p_E, \gamma)$ .

(i) *Strong complementarity.* Then  $\Delta_L / \Delta_H < q_{LE} / q_{HE} < 1 \leq \frac{1 - Q_L \gamma}{1 - Q_H \gamma}$ , so  $\frac{\partial P_H^f}{\partial p_E} < 0$ . Together with  $\frac{\partial P_H^f}{\partial \gamma} < 0$  and  $v'' \geq 0$ , the second term of (11) is nonnegative. The analogue of Lemma 2 (ii) gives  $\frac{\partial U(p_E, \gamma)}{\partial p_E} > 0$ , so the first term is positive. Hence  $\frac{\partial^2 U(p_E, \gamma)}{\partial p_E \partial \gamma} > 0$ .

(ii) *Strong substitutability.* Then  $\Delta_L / \Delta_H > \frac{1 - q_{LE}}{1 - q_{HE}} \geq \frac{1 - Q_L \gamma}{1 - Q_H \gamma}$ , so  $\frac{\partial P_H^f}{\partial p_E} > 0$ . Together with  $\frac{\partial P_H^f}{\partial \gamma} < 0$ , the second term of (11) is nonpositive, and the analogue of Lemma 2 (iii) gives  $\frac{\partial U(p_E, \gamma)}{\partial p_E} < 0$ . Hence  $\frac{\partial^2 U(p_E, \gamma)}{\partial p_E \partial \gamma} < 0$ .  $\square$

*Proof of Proposition 5.* Let us define the second-period reputation of the reappointed manager for the project whose ease probability is  $p_E$  as follows.

$$P_H^{o1s}(p_E) = \frac{P_H^{o1}(p_E q_{HE} + (1 - p_E) q_{HD})}{P_H^{o1}(p_E q_{HE} + (1 - p_E) q_{HD}) + (1 - P_H^{o1})(p_E q_{LE} + (1 - p_E) q_{LD})}$$

$$P_H^{o1f}(p_E) = \frac{P_H^{o1}(1 - p_E q_{HE} - (1 - p_E) q_{HD})}{P_H^{o1}(1 - p_E q_{HE} - (1 - p_E) q_{HD}) + (1 - P_H^{o1})(1 - p_E q_{LE} - (1 - p_E) q_{LD})}$$

Let  $p_E^{o1} = P_E^{o1}(p_E^0)$ . Then, the expected utility of accepting the reappointment offer is  $\mu + U_2^{o1}(p_E^{o1})$ ,

where

$$U_2^{o_1}(p_E) = Q(p_E, P_H^{o_1})v(P_H^{o_1s}(p_E)) + (1 - Q(p_E, P_H^{o_1}))v(P_H^{o_1f}(p_E)).$$

Similar to Lemma 2, we can show that  $U_2^{o_1}$  is increasing under the complementarity case and decreasing in the substitutability case. In contrast, the reservation utility, the utility of managing a project with task ease  $p_E^0$ , is  $v(p_H^{o_1}) + U_2^{o_1}(p_E^0)$ . Therefore, to employ the current manager, the firm needs to pay

$$\mu = U_2^{o_1}(p_E^0) - U_2^{o_1}(P_E^{o_1}) + v(P_H^{o_1}(p_E^0)).$$

Compared with employing a worker as a manager, the benefit of reappointing the current manager over employing a manager with having reputation  $p_H$  for the firm is

$$\begin{aligned} R_s &= \Pi(P_H^s - p_H)(P_E^s - p_E^0)(\Delta_H - \Delta_L) + U_2^s(P_E^s) - U_2^s(p_E^0) \\ R_f &= \Pi(P_H^f - p_H)(P_E^f(q_{HE} - q_{LE}) + (1 - P_E^f)(q_{HD} - q_{LD})) + U_2^f(P_E^f) - U_2^f(p_E^0). \end{aligned}$$

To calculate the above equations, we use the specification:

$$v(p_H') = \max\{(p_H' - p_H), 0\} \cdot \Pi \cdot [p_E^0 \cdot (q_{HE} - q_{LE}) + (1 - p_E^0) \cdot (q_{HD} - q_{LD})].$$

We have two cases.

- (i) Consider the complementarity case ( $\Delta_L/\Delta_H < \frac{q_{LE}}{q_{HE}}$ ). Note that this assumption implies that  $\Delta_H > \Delta_L$ . By  $P_E^s > p_E^0$ ,  $U_2$  is increasing in  $p_E$  and  $\Delta_H > \Delta_L$ ,  $R_s > 0$ . Therefore, the reappointment of the current manager is indeed more beneficial than employing a new worker. If  $o_1 = f$ , the gain from reappointing the current manager is  $R_f < 0$  as  $P_H^f < p_H$ , and  $P_E^f < p_E^0$ . Therefore, if the firm does not promote the current worker, it hires an external worker.
- (ii) Consider the substitutability case ( $\Delta_L/\Delta_H > \frac{1-q_{LE}}{1-q_{HE}}$ ). This implies that  $\Delta_L > \Delta_H$  and  $U_2$  is decreasing. Then,  $R_s < 0$ . To determine the sign of  $R_f$ , we consider  $P_H^f(p_E^0)$ . We will show

that  $P_H^{fs}(p_E^0) < p_H$  under our assumption, which implies that  $U_2 = 0$ . Note that

$$\frac{P_H^{fs}(p_E^0)}{1 - P_H^{fs}(p_E^0)} = \frac{P_H^f}{1 - P_H^f} \frac{p_E^0 \Delta_H + q_{HD}}{p_E^0 \Delta_L + q_{LD}} = \frac{p_H}{1 - p_H} \frac{p_E^0 \Delta_H + q_{HD}}{p_E^0 \Delta_L + q_{LD}} \frac{(1 - q_{HD}) - p_E^0 \Delta_H}{(1 - q_{LD}) - p_E^0 \Delta_L}.$$

Furthermore,

$$\begin{aligned} & \frac{p_E^0 \Delta_H + q_{HD}}{p_E^0 \Delta_L + q_{LD}} \frac{(1 - q_{HD}) - p_E^0 \Delta_H}{(1 - q_{LD}) - p_E^0 \Delta_L} < 1 \\ \iff & (p_E^0)^2 (\Delta_L^2 - \Delta_H^2) + ((1 - 2q_{HD})\Delta_H - (1 - 2q_{LD})\Delta_L)p_E^0 + q_{HD}(1 - q_{HD}) - q_{LD}(1 - q_{LD}) < 0. \end{aligned}$$

As  $\Delta_L^2 > \Delta_H^2$ , the above quadratic function is maximized at  $p_E \in \{0, 1\}$ . Therefore,

$$\frac{p_E^0 \Delta_H + q_{HD}}{p_E^0 \Delta_L + q_{LD}} \frac{(1 - q_{HD}) - p_E^0 \Delta_H}{(1 - q_{LD}) - p_E^0 \Delta_L} < 1 \text{ if } \max \left\{ \frac{q_{HE}(1 - q_{HE})}{q_{LE}(1 - q_{LE})}, \frac{q_{HD}(1 - q_{HD})}{q_{LD}(1 - q_{LD})} \right\} < 1.$$

By assumption, as  $q_{LD} > 1/2$ ,  $\min\{q_{LD}, q_{HD}, q_{LE}, q_{HE}\} > 1/2$ , and then,  $q_{LD}(1 - q_{LD}) > q_{HD}(1 - q_{HD})$  and  $q_{LE}(1 - q_{LE}) > q_{HE}(1 - q_{HE})$ . Therefore, this implies that  $P_H^{fs}(p_E^0) < p_H$ . Further, as  $P_H^{fs}(p_E^0) \geq \max\{P_H^{ff}(p_E), P_H^{fs}(P_E^f), P_H^{ff}(P_E^f)\}$ ,  $U_2^f(P_E^f) = U_2^f(p_E^0) = 0$ . Therefore,  $R_f < 0$ .

□

*Proof of Proposition 7.* We examine when  $U'(p_E) < 0$ . Note that

$$\begin{aligned} U'(p_E) &= Q'(p_E)[v^s - v^f] + Q(p_E)v^{s'}P^{s'} + (1 - Q(p_E))v^{f'}P^{f'}, \\ \text{where } v^o &= v(P_H^o(p_E)), v^{o'} = v'(P_H^o(p_E)), P^{o'} = \frac{\partial P_H^o(p_E)}{\partial p_E}, \\ &\text{for each } o \in \{s, f\}. \end{aligned}$$

By the convexity of  $v$  and  $v''' \geq 0$ , we can show that  $\frac{v'(a) + v'(b)}{2}(a - b) \geq v(a) - v(b)$ . Then,

$$U'(p_E) \leq \frac{P^s - P^f}{2} Q'(p_E)(v^{s'} + v^{f'}) + Q(p_E)v^{s'}P^{s'} + (1 - Q(p_E))v^{f'}P^{f'},$$

where  $P^o = P_H^o(p_E)$  for each  $o \in \{s, f\}$ . By the law of iterated expectation,  $p_H = Q(p_E)P_H^s(p_E) +$

$(1 - Q(p_E))P_H^f(p_E)$ , and therefore, by a differentiation,

$$Q'(p_E)[P^s - P^f] = -[Q(p_E)P^{s'} + (1 - Q(p_E))P^{f'}]. \quad (12)$$

Then,

$$U'(p_E) \leq (v^{s'} - v^{f'}) \frac{1}{2} [Q(p_E)P^{s'} - (1 - Q(p_E))P^{f'}].$$

By using the formula shown in the proof of Lemma 1 ((vi)),

$$[Q(p_E)P^{s'} - (1 - Q(p_E))P^{f'}] = p_E p_H p_L \frac{(Q + Q_L - 2QQ_L)\Delta_H - (Q + Q_H - 2QQ_H)\Delta_L}{Q(1 - Q)}. \quad (13)$$

Then, if the following inequality holds,  $U'(p_E) < 0$ .

$$\frac{\Delta_L}{\Delta_H} > \frac{Q + Q_L - 2QQ_L}{Q + Q_H - 2QQ_H}.$$

We show the reverse inequality. Note that by  $\frac{v'(a)+v'(b)}{2}(a-b) \geq v(a) - v(b)$ ,

$$\begin{aligned} U'(p_E) &= Q'(p_E)[v^s - v^f] + Q(p_E)v^{s'}P^{s'} + (1 - Q(p_E))v^{f'}P^{f'}, \\ &\geq Q'(p_E)[v^s - v^f] + Q(p_E)v^{s'}P^{s'} + (1 - Q(p_E)) \left[ 2 \frac{(v^s - v^f)}{P^s - P^f} - v^{s'} \right] P^{f'} \\ &= \left[ Q'(p_E) + 2P^{f'} \frac{1 - Q(p_E)}{P^s - P^f} \right] [v^s - v^f] + [Q(p_E)P^{s'} - (1 - Q(p_E))P^{f'}] v^{s'} \end{aligned}$$

By (12),

$$\begin{aligned} \left[ Q'(p_E) + 2P^{f'} \frac{1 - Q(p_E)}{P^s - P^f} \right] &= \frac{1}{P^s - P^f} [Q'(p_E)(P^s - P^f) + 2P^{f'}(1 - Q(p_E))] \\ &= \frac{-1}{P^s - P^f} [Q(p_E)P^{s'} - P^{f'}(1 - Q(p_E))] \end{aligned}$$

Therefore,

$$U'(p_E) \geq [Q(p_E)P^{s'} - (1 - Q(p_E))P^{f'}] \left[ v^{s'} - \frac{v^s - v^f}{P^s - P^f} \right]$$

By the convexity,  $v^{s'} \geq \frac{v^s - v^f}{p^s - p^f}$ . Then, by (13), if  $\frac{\Delta_L}{\Delta_H} < \frac{Q + Q_L - 2QQ_L}{Q + Q_H - 2QQ_H}$ ,  $U'(p_E) > 0$ . □

## A.1 Proofs of Propositions 2 and 6

Proofs of Propositions 2 and 6 follow a similar argument to the proof of Proposition 1, and thus, we relegate the proofs to the supplementary materials.

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## **Declaration of generative AI and AI-assisted technologies in the writing process**

During the preparation of this work, the authors used Grammarly, ChatGPT and Claude to obtain writing assistance. After using this tool/service, the authors reviewed and edited the content as

needed and took full responsibility for the publication's content.



# Supplementary Materials for Promotion, Continuation, and Reputational Volatility under Task-Ease Uncertainty

Kohei Yamagata and Tomoya Tajika

This material provides the proofs of Proposition 6 and Proposition 2, which are omitted in the main text.

*Proof of Proposition 2.* Recall that

$$\tilde{B} = \frac{1}{Q(p_E^0)} \left[ c - Q(p_E^0) \left[ \pi^s [U(P_E^s, e_2^s) - U(p_E^0)] - \pi^f [U(P_E^f, e_2^f) - U(p_E^0)] \right] + \pi^f [U(p_E^0, e_2^f) - U(P_E^f, e_2^f)] \right].$$

- (i) Suppose that  $\Delta_L/\Delta_H < \frac{q_{LE}}{q_{HE}}$ . From Lemma 2 (ii),  $U$  is increasing in  $p_E$ . Then, as  $P_E^s > p_E^0$ ,  $U(P_E^s, 1) \geq U(p_E^0) > U(P_E^s, 0) = v(p_H)$ , which implies that  $\pi^s = 1$  is optimal if  $e_2^s = 1$  and  $\pi^s = 0$  is optimal if  $e_2^s = 0$ . Also,

$$Q(p_E^0) \frac{\partial \tilde{B}}{\partial \pi^f} = Q(p_E^0) \underbrace{[U(P_E^f, e_2^f) - U(p_E^0)]}_{\text{negative}} + \underbrace{U(p_E^0, e_2^f) - U(P_E^f, e_2^f)}_{\text{positive}}.$$

Then, to minimize  $\tilde{B}$ ,  $\pi^f = 0$  is optimal if  $e_2^f = 1$ , and  $\pi^f = 1$  is optimal if  $e_2^f = 0$ . Note that as the project ends in period 2, we need not consider the effect of reputation in the incentive pay  $(\beta_2^s, \beta_2^f)$ . Therefore, the optimal payment to incentivize the effort of the second-period worker is  $\beta_2^o = c/Q(P_E^o)$  for each  $o \in \{s, f\}$ .

By using the relation  $\pi^s = e_2^s$  and  $\pi^f = 1 - e_2^f$ , the total profit is

$$\begin{aligned} & \left[ Q(p_E^0) [Q(P_E^s) \Pi - c] e_2^s + (1 - Q(p_E^0)) [Q(P_E^f) \Pi - c] e_2^f \right] \\ & + Q(p_E^0) [\Pi + [U(P_E^s) - U(p_E^0)] e_2^s] - c + Q(p_E^0) [U(p_E^0) - v(p_H)] (1 - e_2^f). \end{aligned}$$

Then,  $e_2^s = 1$  is optimal as  $Q(P_E^s)\Pi - c > Q(p_E^0)\Pi - c > 0$ .  $e_2^f = 1$  if and only if  $(1 - Q(p_E^0))[Q(P_E^f)\Pi - c] > Q(p_E^0)[U(p_E^0) - v(p_H)]$ .

Finally, we check the optimality to enforce the first period worker to effort. If  $e_1 = 0$ . Then, the total profit is  $[Q(p_E^0)\Pi - c]$ .  $e_1 = 1$  is optimal if

$$\begin{aligned} & \left[ Q(p_E^0)[Q(P_E^s)\Pi - c] + \left[ (1 - Q(p_E^0))[Q(P_E^f)\Pi - c] - Q(p_E^0)[U(p_E^0) - v(p_H)] \right] e_2^f \right] \\ & + Q(p_E^0)[U(P_E^s) - U(p_E^0)] + Q(p_E^0)[U(p_E^0) - v(p_H)] > 0, \end{aligned}$$

which is satisfied.

(ii) Suppose that  $\Delta_L/\Delta_H > \frac{1-q_{LE}}{1-q_{HE}}$ . Then, as  $U(P_E^s, e_2^s) \leq U(p_E^0)$ ,  $\pi^s = 0$  is optimal. Also,

$$Q(p_E) \frac{\partial B}{\partial \pi^f} = Q(p_E^0) \underbrace{[U(P_E^f, e_2^f) - U(p_E^0)]}_{\text{positive iff } e_2^f=1} + \underbrace{U(p_E^0, e_2^f) - U(P_E^f, e_2^f)}_{\text{negative}}$$

Then, if  $e_2^f = 1$ ,  $Q(p_E^0) \frac{\partial B}{\partial \pi^f} = [U(p_E^0) - U(P_E^f)](1 - Q(p_E^0)) < 0$ . On the contrary, if  $e_2^f = 0$ ,  $Q(p_E^0) \frac{\partial B}{\partial \pi^f} = Q(p_E^0)[v(p_H) - U(p_E^0)] < 0$ . In any case,  $\pi^f = 1$  is optimal. Now the total profit is

$$\begin{aligned} & \left[ Q(p_E^0)[Q(P_E^s)\Pi - c]e_2^s + (1 - Q(p_E^0))[Q(P_E^f)\Pi - c]e_2^f \right] \\ & + Q(p_E^0)\Pi + (1 - Q(p_E^0))U(P_E^f, e_2^f) + Q(p_E^0)U(p_E^0) - U(p_E^0, e_2^f) - c \end{aligned}$$

Then,  $e_2^s = 1$  is optimal.  $e_2^f = 1$  is optimal if

$$(1 - Q(p_E^0))[Q(P_E^f)\Pi - c] > \xi(p_E^0).$$

$e_1 = 1$  is optimal if

$$\begin{aligned} & \left[ Q(p_E^0)[Q(P_E^s)\Pi - c] + (1 - Q(p_E^0))[Q(P_E^f)\Pi - c]e_2^f \right] \\ & + (1 - Q(p_E^0))[U(P_E^f, e_2^f) - U(p_E^0, e_2^f)] + Q(p_E^0)[U(p_E^0) - U(p_E^0, e_2^f)] > 0, \end{aligned}$$

which is satisfied.

□

*Proof of Proposition 6.* Recall that

$$B^\rho := \frac{1}{Q(p_E^0)} \left[ c - Q(p_E^0) \left[ \pi^s [U(P_E^s, e_2^s) - v(p_H)] - \pi^f [U(P_E^f, e_2^f) - v(p_H)] \right] \right. \\ \left. + \rho \pi^f [U(p_E^0, e_2^f) - U(P_E^f, e_2^f)] \right].$$

(i) Suppose that  $\Delta_L/\Delta_H < \frac{q_{LE}}{q_{HE}}$ . Then, as  $U(P_E^s, e) \geq v(p_H)$ ,  $\pi^s = 1$  is optimal. Also,

$$Q(p_E^0) \frac{\partial B^\rho}{\partial \pi^f} = Q(p_E^0) \underbrace{[U(P_E^f, e_2^f) - v(p_H)]}_{\text{positive}} + \rho \underbrace{[U(p_E^0, e_2^f) - U(P_E^f, e_2^f)]}_{\text{positive}}$$

Then, independent of  $e_2^f$ ,  $\pi^f = 0$  is optimal. Note that as the project ends in period 2, we need not consider the effect of reputation in the incentive pay  $\beta_2$ . Therefore, the payment for the second-period worker is  $\beta_2 = c/Q(P_E^o)$ . Now the total profit is

$$\left[ Q(p_E^0) [Q(P_E^s) \Pi - c] e_2^s + (1 - Q(p_E^0)) [Q(P_E^f) \Pi - c] e_2^f \right] \\ + Q(p_E^0) [\Pi + U(P_E^s, e_2^s) - U(p_E^0)] - c$$

Then,  $e_2^s = 1$  is optimal as  $Q(P_E^s) \Pi - c > Q(p_E^0) \Pi - c > 0$ .  $e_2^f = 1$  if and only if  $(1 - Q(p_E^0)) [Q(P_E^f) \Pi - c] > 0$ .

(ii) Suppose that  $\Delta_L/\Delta_H > \frac{1-q_{LE}}{1-q_{HE}}$ . Then, as  $U(P_E^s, e_2^s) > v(p_H)$ ,  $\pi^s = 1$  is optimal. Also,

$$Q(p_E) \frac{\partial B}{\partial \pi^f} = Q(p_E) \underbrace{[U(P_E^f, e_2^f) - v(p_H)]}_{\text{positive}} + \rho \underbrace{[U(p_E^0, e_2^f) - U(P_E^f, e_2^f)]}_{\text{negative}}$$

Then, if  $e_2^f = 1$ ,  $\frac{\partial B}{\partial \pi^f} < 0$  if and only if

$$\frac{Q(p_E)}{\rho} < \frac{U(P_E^f) - U(p_E^0)}{U(P_E^f) - v(p_H)}. \quad (14)$$

On the contrary, if  $e_2^f = 0$ ,  $\frac{\partial B}{\partial \pi^f} = 0$ . This implies that  $\pi^f = 1$  if and only if eq. (14) holds in the optimal contract.

Now the total profit is

$$\begin{aligned} & \left[ Q(p_E^0)[Q(P_E^s)\Pi - c]e_2^s + (1 - Q(p_E^0))[Q(P_E^f)\Pi - c]e_2^f \right] + Q(p_E^0)\Pi - c \\ & + Q(p_E^0)(U(P_E^s) - v(p_H)) + [Q(p_E^0)(U(P_E^f) - v(p_H)) - \rho(U(P_E^f) - U(p_E^0))]\pi^f e_2^f. \end{aligned}$$

Then,  $e_2^s = 1$  is optimal.  $e_2^f = 1$  is optimal if

$$(1 - Q(p_E^0))[Q(P_E^f)\Pi - c] + \min\{Q(p_E^0)(U(P_E^f) - v(p_H)) - \rho(U(P_E^f) - U(p_E^0)), 0\} > 0.$$

□